

Equilibrium Provision of Information in Two-Sided Markets: Evidence from an Online Job Platform

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Abstract

This paper investigates the trade-off agents face when providing information to potential matches in two-sided markets. Specifically, we explore how, in online labor markets, market congestion can lead to under-provision of information from workers to employers and ultimately negatively impacting match efficiency. We develop a model of equilibrium information provision characterized by attention-constrained employers and workers who provide match-specific costly information to their potential employers. The model shows that congestion forms a binding attention constraint, leading to under-provision of signaling effort by workers in equilibrium. We then provide empirical evidence from a major task-based online labor market that supports our theoretical model, showing that congestion leads to hasty and generic applications and that the market is characterized by low job-filling rates. Our model and setting lay the groundwork needed to quantify the loss of efficiency that arises due to information under-provision in congested markets.

Keywords: Digital platforms, Two-sided matching markets, Trade in Services, Freelancing, Congestion, Communication Efforts

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1. Introduction

Finding a “match” in a two-sided market, whether it is an employer looking for a worker, a tenant looking for a house, or a student looking for a school, requires agents on at least one side to provide information about themselves to agents on the other side. While some of this information can be publicly observable, a key mechanism by which agents find high-value matches is by providing information about themselves that is specific to the potential match of interest. Providing such information can be costly. For example, in addition to submitting a generic CV, job searchers often spend time crafting a job-specific application tailored to a job post that signals they will be a good fit. These information provision costs present agents with a trade-off: Is it worth expending the effort to signal match-specific information about themselves, even when that match may not form?

In this paper, we study how agents navigate this information provision trade-off in equilibrium when competing for the same matches. In particular, we study how market congestion—when agents are overwhelmed with too many potential matches to consider them all—can lead agents to under-provide information in equilibrium, eroding match efficiency.

We explore these topics in the context of task-based online labor markets, an ideal and increasingly economically relevant setting (Autor, 2001) in which to study the information provision trade-off under congestion. Task-based online labor markets are particularly sensitive to the information provision trade-off. Because jobs listed on task-based online labor markets tend to be one-time isolated tasks, the cost of providing information (as an applicant) or of screening applicants (as an employer) is a larger fraction of the overall value of a hire relative to the in-person salaried labor market, thus raising the potential for information provision costs to impede efficient hiring. Yet information provision is essential to screening in these markets: because employers must commit to pay workers who they will never meet in person and will likely only have limited contact with, the onus is on applicants to provide information about themselves to their potential employers to convince them of their match quality with the job. Furthermore, in the presence of congestion, competition for each job increases, potentially leading workers to be less willing to spend time personalizing applications for each job. In equilibrium, all workers may under-provide information by writing less specific job applications, eroding information transmission between workers and employers. The result is that employers may have low levels of trust in the applications they receive and leave their job posts unfilled, lowering even further the chances that workers costly provision of information will pay off.

To concretely establish the interaction between congestion and the information provision trade-off, we develop a model for equilibrium information provision between employers and workers on an online labor market in which employers are attention constrained and workers need to send match-specific costly information to convince employers to hire them. We build a static model of job matching in which many workers exogenously arrive at a job posting and send applications including a price bid and a signal of match quality. Employers then look through applications and choose a single worker or the outside option to maximize their expected payoffs.

Worker payoffs are only sensitive to their own opportunity costs of supplying their labor,

the amount they will be paid, and the the effort costs of signaling, while employers only care about the match quality of the workers they hire and the amount they pay them. While workers know their opportunity costs before applying, we make the crucial assumption that, conditional on observables, workers do not know their match value with the jobs for which they apply. The idea behind this assumption is that workers know exactly how costly it will be to supply their labor for a specific task, but they do not know exactly how the employers will value that labor once it has been completed. This informational assumption changes the model from one in which workers endeavor to share their private information about their type to one in which workers choose to expend costly effort to send signals, the output of which will partially reveals their types (Holmström, 1999). This assumption allows for a tractable signaling model where workers only have one dimensional-private information – their costs – and employers do not form preferences over those private types. This part of the model encapsulates how workers can expend costly effort to provide information to employers.

The next key aspect of our model is that employers are attention constrained. They are not able to consider all of the applications they receive, and they rely on the platform to order applications for them according to a recommender system that only takes into account observable characteristics of workers, rather than their bids or signals. Workers do not know for sure whether their applications will be considered, but they form accurate beliefs over their chances of being considered and factor this into their information provision decisions. Thus, we show how congestion in the form of binding attention constraints leads to under provision of signaling effort in equilibrium and how this can lead the market to have inefficiently low job-filling rates.

We then elucidate how our model is empirically tractable and sketch an argument for how the model could be identified with data from an online labor market.¹

Finally, we use data from Freelancer.com, a major international online labor market, to present descriptive evidence of this hypothesized equilibrium in which congestion leads to under provision of information from workers to employers. On the employers' side, we find that employers have to sort through many undifferentiated bids, many of which have been hastily written and poorly customized. These findings paired with the fact that, in our sample, only 40% of posted jobs get filled, suggests that workers are not providing enough information to convince employers to hire them, and as employers have to look through more and more of these insufficient applications, they opt for their outside options. On the workers' side, we find that workers submit applications soon after projects are posted and measure that they spend very little time reading the job post and submitting their applications. These findings paired with the fact that, in our sample, only 10% of workers ever find jobs on the platform, and the acceptance rate per application is quite low for both workers with and without reputations on the platform lends suggestive evidence that workers internalize their low chances of consideration and accordingly spend little effort on applications. In sum, these findings are suggestive evidence for the forces highlighted in our model and suggest that future work should be done to quantify how much efficiency is lost to under-provision of information in equilibrium under congestion in this market.

¹We plan to estimate this model in a follow-up paper.

Our paper builds most directly upon two strands of literature – one that studies congestion as a friction in market design (Roth, 1982; Roth and Peranson, 1999; Roth, 2008; Ashlagi et al., 2020; Arnosti, Johari, and Kanoria, 2021), and another that focuses on empirical studies of online labor markets (Agrawal et al., 2015; Barach, Golden, and Horton, 2020; Brinatti et al., 2021; Krasnokutskaya, Song, and Tang, 2020; Marinescu and Wolthoff, 2020) – both of which have grown considerably in recent years as online labor markets and online two-sided markets in general have expanded (Kässi, Lehdonvirta, and Stephany, 2021). *To be expanded on.*

The rest of the paper proceeds as follows. Section 2 describes how our market of focus, Freelancer.com, operates, and the data that we use. Section 3 presents our equilibrium model of information provision. Section 4 builds upon the prior two sections by presenting reduced-form evidence of the model implications in the data. Finally, Section 5 concludes.

2. Market Description and Data

This paper uses new data from Freelancer.com, a major online labor market. The first part of this section describes the market structure, including the platform’s players, the type of tasks outsourced, and the job matching process. The second part details the structure of the data we obtained through our collaboration with Freelancer.com.

2.1. Market Description: A Global Digital Marketplace

Freelancer.com is one of the world’s largest web-based job marketplace by number of users and posted jobs. Digital marketplaces for freelance labor, such as Freelancer.com or Upwork, match global workers with potential employers around the world to trade tasks that are delivered remotely, such as software development, sales and marketing support, and creative work - what is called “freelancing”. Since its creation in 2009, more than 71 million users from over 247 countries, regions and territories signed up for remote work on this globalized market. In fiscal year 2019 alone, over USD 123 million in gross value of payments were made through the platform. With an average payment per job of USD 190, this means that 650,000 job tasks were completed on the platform on average in 2019.

2.1.1. A North-South Trade in Tasks

Digital marketplaces globalize traditionally local labor markets, with platforms that enable employers primarily located in high-income countries to easily outsource tasks to contractors, primarily located in low-income countries. They also provide workers in the Global South with work opportunities that may not have been offered in their local labor markets. Freelancer.com explicitly offers low barriers to entry for workers (a main difference with competing platforms like Upwork). The outsourcing in tasks on this market is thus characterised by a North-South exchange, with most employers located in the Global North and most workers in the Global South.

Figure 1 compares the geographical distribution of workers and employers on the platform. The top panel displays the top 10 countries of employers. 50% of all jobs are posted from employers located in the United States (25.5%), Australia (11%), the United Kingdom (9.5%) and Canada (5%). While India also appears to be a large poster of jobs on the platform (8.5%), it is above all the primary country of origin of workers: 28% of all freelancers are from India. Pakistan (7%), the Philippines (5%), Indonesia (4%) and Bangladesh (3%) follow.

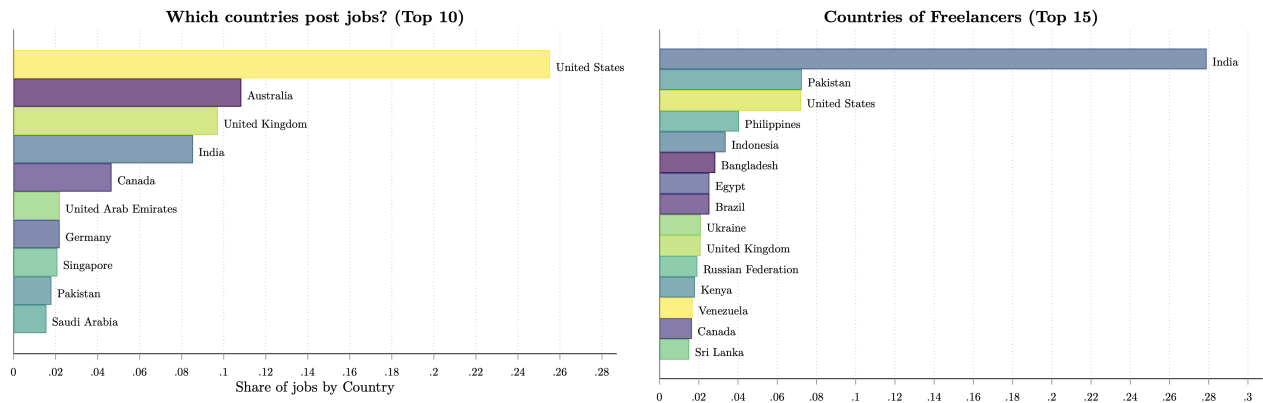


Figure 1: Who is on the platform?

Notes: These figures plot the listed country of workers (“freelancers”, on the right) and employers (individuals or firms posting jobs, on the left). Firms on the platform are largely from the Global North, while freelancers on the platform are mostly from the global south

2.1.2. Types of Jobs Posted

Online labor markets combine two increasingly important pillars of the modern economy. First, they are digital platform markets — marketplaces in which a third party facilitates matches between buyers and sellers. Second, they are largely centered around gig work — tasks that firms or individuals outsource to independent workers. These jobs can be done remotely and require little capital other than a computer. Workers on Freelancer.com are mostly hired to perform jobs in areas such as software development, writing, data entry and design (Figure 2). Each broad occupation can be further disaggregated into detailed occupations and skills. For a detailed list of all possible skills and job categories, see Appendix Table C.1.

2.1.3. How do Employers and Workers Match for Jobs?

Figure 3 describes the two-sided market’s dynamics of the platform. The job matching process can be decomposed into five main steps.

In a first step, firms or individual employers (“clients”), looking to outsource a specific task, first post jobs (“projects”) on the Freelancer.com platform. In a second step, workers with matching skills to those required for the task place bids on the job project. These bids include two components: first, how much they are asking to be paid for the task and

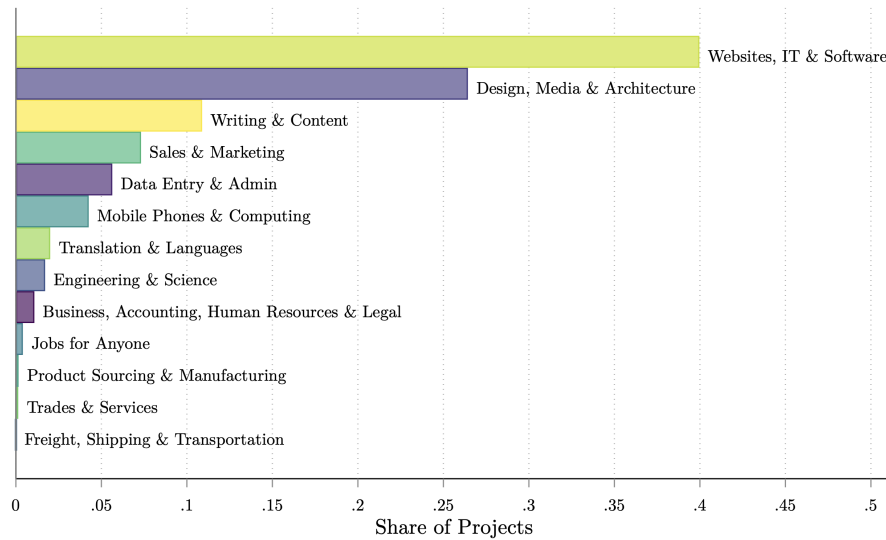


Figure 2: What Types of Jobs Are Posted on the Platform?

Notes: This figure plots the aggregate job categories listed on the platform. Coding, design, and writing skills are most in demand on the platform. For a detailed list of all possible skills and job categories, see Appendix Table C.1

second a written job proposal functioning as a cover letter. In a third step, given the current set of competitive bids and the characteristics of the workers competing for the tasks, the platform ranks bids for employers according to a recommendation algorithm. In a fourth step, employers then choose the worker with the most attractive bid elements to complete the task, and work commences. Employers can also choose the outside option of not picking anyone and leaving the job unfilled. In the fifth and last step, after completion of the job, both the worker and the employer can evaluate the quality of the job done by attributing a “star rating” to the other party.

Several other characteristics of workers’ bids and employers and workers’ profiles are important to underline.

Searching for Jobs. Figure C.1 provides a screenshot of the platform’s appearance from the point of view of workers looking for jobs. Workers can search job postings by budget and budget type (either fixed-price projects or hourly projects), by project duration, by language or by job categories and skills. Within a given job category, jobs are automatically ranked by posting time, with the newest jobs ranked first. Workers can also sort jobs by budget or number of bids/entries. Each job posting comes with a description of the job tasks and timing, a list of skills necessary for the project, as well as the current number of bids for this job and the average bid amount from competitors.

Workers and Employers Information. Before selecting a worker for a job, employers can see the information on the freelancer’s bid card as well as the freelancer’s profile if they click on their name (see Figures C.3 and C.2).² Workers can also see some limited information

²Employers looking for a worker to complete a project can either publicly post a project and open it for competitive bidding, or they can browse through freelancer profiles and directly send a request for hiring to a specific worker (see Figure C.2). However, the posting of a fixed-price project is the vastly predominant mode by which employers look for workers. The “Freelancers for Hire” section was not a prevalent feature of the

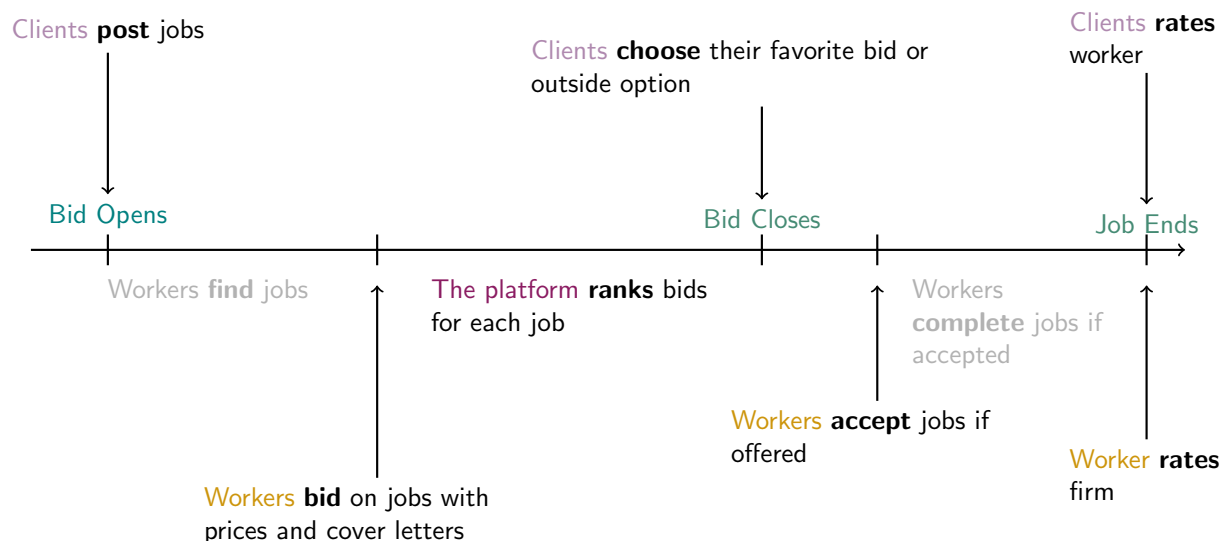


Figure 3: Matching Process

Notes: This figure describes the process during which employers and workers match for jobs on Freelancer.com. In a first step, firms or individual employers (“clients”), looking to outsource a specific task, first post jobs (“projects”) on the Freelancer.com platform. In a second step, workers with matching skills to those required for the task place bids on the job project. These bids include two components: first, how much they are asking to be paid for the task and second a written job proposal functioning as a cover letter. In a third step, given the current set of worker competing for the tasks, the platform ranks bids for employers according to a recommendation algorithm. In a fourth step, employers then choose the worker with the most attractive bid elements to complete the task, and work commences. Employers can also choose the outside option of not picking anyone and leaving the job unfilled. In the fifth and last step, after completion of the job, both the worker and the employer can evaluate the quality of the job don by attributing a “star rating” to the other party.

about the clients when viewing the project details (see Figure C.4).

Accepting a Job. Bids posted by workers during the auction process are not binding. Workers have 46hours to accept or reject a project after being awarded a job from an employer. If no action is taken by the worker within 46hours, the platform considers that this is an implicit reject. Rejections remain relatively rare.

Employers-Workers Communication Tools. Platform-guided recommendations and direct communication are key components of matches. There are no interviews by employers post-acceptation of a project . Before selection however, both employers and workers can communicate in a decentralized, private chat system hosted by the platform.

Upgrades and Memberships. Bidding for a job is free of charges, but there is a finite number of bids you are able to make as a worker. Membership plans allow to increase the number of monthly bids that workers can make (see Figure A.3 for a detailed description of available plans and pricing, and Figure C.5 for a screenshot of the current membership page). Freelancer.com also offers an upgrade system that allows both employers and

workers to improve the visibility and appeal of their projects or proposals through paid features. These upgrades include options for highlighting projects, marking them as urgent, or featuring them on the website’s homepage, thereby increasing the likelihood of attracting higher-quality bids from freelancers. Additionally, freelancers can use upgrades to promote their bids, making them more noticeable to potential clients. Upgrades are tied to a project, not to an employer or a worker.

Payment System. Freelancer.com directly manages secured payment transactions between parties through their milestone payment system (see pictures of payments in Figure C.5). This system operates as an escrow mechanism and ensures mutual trust between clients and freelancers. This arrangement allows funds to be securely held until predefined project milestones are achieved, at which point payments can be released to workers. Such a system not only guarantees financial security for the service provider but also offers clients control over the payment process, contingent on their satisfaction with the work delivered.

2.1.4. The Current Recommender System

Unlike most in-person labor markets, online labor markets are literally designed, which provides a rare opportunity for realistic labor market design. The current recommender system is designed around a deterministic function, primarily focusing on the historical work performance of workers on the platform. The main component of workers’ history that the platform uses to rank freelancer bids is their star rating (see example in Appendix Figure C.3) and their listed job categories and skills expertise. The algorithm is insensitive to the two main components of a worker’s bid: the bid amount and the quality of their job application proposal.

Although this approach rewards freelancers with established track records, placing them at the forefront of bid rankings, it also inherently makes it challenging for newcomers to gain visibility. The ranking algorithm rewards good bidders rather than bad bids, regardless of the merit or competitiveness of their proposals.

2.2. Data

We have access to all data recording activity on the platform. The data we use can be classified into 8 categories. The two main categories of data include job postings (title, description, timestamps, user ID (not identity) of poster, etc.) and job applications, i.e., bids (written bid proposals, dollar value of bid, time value of bid, timestamps measuring how long each worker spends writing their bid proposals, user ID (not identity) of bidder, etc.). We also have information on job completion and payments (which bidder, if any, was selected to complete the project, ex-post ratings of the workers and clients, timestamps, tasks completion status, etc.), and on the work history of jobs completed on the platform. We eventually have data on workers (freelancers) gender and country, on observable workers profiles, and on user-interface data such as clicks data. This last category allows us to compute “consideration sets” for employers (which workers were considered) and workers (which jobs did they consider applying to).

We eventually have access to information on past implementations of certain features of the platform design and other institutional information from `Freelancer.com` on how the platform is designed and run – for example, its bid proposal ranking algorithm, which is the main object of study in our proposed simulation.

Appendix B further details the main variables used in this paper.

3. A Model of Information Provision in a Two-Sided Market

In this section, we present our model of information provision in a two-sided matching market with transfers. This model provides a framework for thinking about how congestion can incentivize workers to send less informative signals of their quality to employers. Moreover, this structural model provides an empirical framework that can be readily applied in future work.

The main contribution of our model is to embed a career concerns model of signaling (Holmström, 1999) into a scoring procurement auction in which employers care both about match quality and price. The key idea here is that we model signaling effort as magnifying the contribution of match quality to the overall signal output relative to the random error. As a result, if congestion incentivizes workers to lower their signaling effort in equilibrium, then congestion will make it more difficult for employers to learn about match quality. With insufficiently high prior beliefs about match quality, congestion will thus lead employers to choose the outside option more frequently, harming the ex-post efficiency of the market.

An outline of the model is as follows. We build a static model of a cross-section of different job postings. A set of workers will randomly arrive at each job, and each of them will apply to the job in the form of a monetary bid and a signal of match quality. Employers will form beliefs about the match quality of each applicant based on the signal, and then choose a single worker or the outside option to maximize their expected utility. Once this choice has been made, the employer pays the chosen worker, or pays no one if the outside option was chosen, and payoffs are realized.

3.1. Environment

Each worker (freelancer) i has a vector of discrete observables $x_i \in X$, including country, verification status, and reputation on the platform. We assume workers have no persistent unobservable types, allowing us to model the matching process as static, omitting time subscripts. Each job (project) j has a vector of discrete observables $w_j \in W$, including country of the employer (client), budget, and skills required. Note that we use j to index both employers and jobs, since we consider them one in the same.

A set of workers, I_j , simultaneously and randomly arrives at job j , where I_j is a $|X|$ -dimensional vector whose entries are the count of each worker of type $x \in X$ matched

to j . Formally, $I_j \in \mathbb{N}^{|\mathcal{X}|}$, and $I_j \sim i.i.d. F_{X,W}(\cdot, w_j)$, where $F_{X,W} : (\mathbb{N}^{|\mathcal{X}|}, W) \rightarrow [0, 1]$ such that for all $w \in W$, $\sum_{I \in \mathbb{N}^{|\mathcal{X}|}} F_{X,W}(I, w) = 1$. We assume that each worker matched to job j will submit an application.

Each worker will have a match quality with the job η_{ij} , which will be unknown to them and drawn i.i.d. according to CDF $F_\eta(\cdot, x_i, w_j)$, which we assume to be $N(m_\eta(x_i, w_j), V_\eta(x_i, w_j))$. The idea behind this assumption is that workers do not know exactly how their work would be valued by the employer, but they do have a correct prior belief distribution based on both their and the job's observable characteristics.

Each employer j receives ex-post utility from choosing bidder i :

$$U_{ij} = \beta(w_j) \cdot \eta_{ij} - \alpha(w_j) \cdot b_{ij} + v_{ij}, \quad (1)$$

where b_{ij} is worker i 's bid on job j , i.e. the payment j commits to pay i if j chooses i , v_{ij} is an idiosyncratic taste shock, known to j but unknown to i , drawn i.i.d. according to CDF $F_v(\cdot)$, which we assume to be a T1EV distribution, and both taste parameters $\beta(\cdot)$ and $\alpha(\cdot)$ are allowed to vary by employer observables w_j . Should an employer j choose the outside option, indexed by $i = 0$, she will receive utility v_{0j} , which will also be drawn i.i.d. from a T1EV distribution. Note that this normalizes the mean value of the outside option.

Employers are attention-constrained and are often unable to consider all applications they receive. Instead, they engage in simultaneous search where they draw an attention constraint $\phi_j \sim i.i.d. F_\phi(\cdot, w_j)$, with $\phi_j \in \mathbb{N}$, and each employer j is only able to consider ϕ_j applications.

The platform orders applications for employers according to a ranking algorithm which dictates which subset of applications employers will consider. Each worker i will have a rank score $r_i \equiv r(x_i) + \zeta_i \in \mathbb{R}$, where $r(\cdot)$ is a real-valued deterministic function, mostly determined by workers' history on the platform, and ζ_i is a random component with a small variance used for tie-breaking, drawn according to CDF $F_\zeta(\cdot)$.³ We assume workers know $r(x_i)$ but not ζ_i . Each application's realized rank $R_{ij} \in \mathbb{N}$ is then computed as an ascending count starting from 1 such that $R_{ij} \leq R_{i'j}$ if and only if $r_{ij} \geq r_{i'j}$. For example, if $R_{ij} = 4$, then i had the fourth highest rank score of all workers applying to job j .

Each employer j will thus consider an application from worker i if and only if $R_{ij} \leq \phi_j$. Note that R_{ij} does not provide any information to employer j conditional on observing x_i . The employer thus only uses the ranking to form consideration sets and not to learn anything about η_{ij} . Also note that workers do not know whether their applications will be considered since they do not know their exact rank score, the rank scores of their competitors, or the attention constraint of the employer. Workers, do, however form beliefs

³In reality, the algorithm is deterministic, but we model it with randomness to capture two features of the data that we omit from the model. First, we are binning some continuous variables in X so that each x_i is a vector of discrete variables. We argue that the small differences we don't capture here by binning continuous variables are effectively random and do not contain relevant information. Second, there are some rarely used variables that go into the rank score that we do not include in our model such as measures of trust that are mostly used to punish left-tail bad actors on the platform, e.g. those who have signed up for multiple accounts with the same IP address.

over their chance of being considered, which is a crucial component of our model we will discuss in more detail later in this section.

When each worker i applies to project j , she will choose a signaling effort $e_{ij} \in (0, \infty)$ at a convex cost $\psi(\cdot, x_i, w_j)$, where for all x and w , $\psi(\cdot, x, w)$ is a twice differentiable, increasing, and convex real-valued function with $\psi(0, x, w) = 0$. The signal that each worker i sends to employer j will also come with random noise ϵ_{ij} unknown to the worker and drawn i.i.d. according to CDF $F_\epsilon(\cdot)$, which we assume to be $N(0, V_\epsilon)$. Signaling effort e_{ij} and signaling noise ϵ_{ij} will come together to produce the signal y_{ij} :

$$y_{ij} = e_{ij} \cdot \eta_{ij} + \epsilon_{ij}. \quad (2)$$

Workers will thus choose e_{ij} in order to imperfectly choose y_{ij} , which will be what the employer uses to form a belief about the pay-off relevant match quality η_{ij} .⁴

If a worker i were to be selected and thus complete a project j , they would incur cost c_{ij} drawn i.i.d. according to CDF $F_c(\cdot, x_i, w_j)$, where $F_c(\cdot, x_i, w_j)$ is absolutely continuous with density $f_c(\cdot, x_i, w_j)$ and support $[\underline{c}_j, \bar{c}_j] \subset \mathbb{R}^+$. Note that we let the bounds of these support vary by j only insofar as they vary with w_j . This cost c_{ij} is known to workers, unknown to employers, conditionally independent of match quality η_{ij} , and independent of ϵ_{ij} . Thus, c_{ij} is the only private information i has about her type that the employer does not know. The motivation behind this assumption is that workers indeed have an exact sense of what they would do to complete the posted job, but they do not know how this work will be received.

After drawing c_{ij} , each worker i applying to j chooses a bid $b_{ij} \in (0, \infty)$ along with her signaling effort e_{ij} . In reality, included in w_j is both a minimum and maximum bid allowed, $\underline{b}_j$ and $\bar{b}_j$ respectively, so each $b_{ij} \in [\underline{b}_j, \bar{b}_j]$. We assume that the range of possible costs, $[\underline{c}_j, \bar{c}_j]$ is such that the bounds $\underline{b}_j$ and $\bar{b}_j$ do not bind in equilibrium. This simplifies our analysis and accords with our assumption that all workers who arrive at job j submit an application. Future work could consider binding bid limits and a selection correction for those who arrive but do not submit applications.

Thus, each freelancer will choose e_{ij} and b_{ij} to maximize:

$$P(j \text{ chooses } i | e_{ij}, b_{ij}, x_i, w_j) \cdot (b_{ij} - c_{ij}) - \psi(e_{ij}, x_i, w_j). \quad (3)$$

Note that workers do not observe their competitors when they arrive at a job posting, and they only observe their own cost c_{ij} and their own observables x_i , and the job's observables w_j . Therefore, the probability function in their expected utility will account for the workers' uncertainty over their competitors' observables, their competitors' unobservables, their own unobserved match qualities, the employer's attention constraints, and all possible draws of signaling error and employer taste shocks. Of course, this probability will also depend on the equilibrium play of all other players, which will be explored in the next subsection.

⁴In future work, we suggest measuring signaling effort by the amount of time freelancers spend interacting with a job post before submitting an application, and we suggest measuring signals by a measure of writing quality that incorporates how well the written portion of the application responds to the job description.

Returning to the employer's problem, we note that employer j will not observe η_{ij} and will only observe i 's bid b_{ij} and i 's signal y_{ij} . Thus, employer j 's ex-ante utility from choosing worker i will be:

$$EU_{ij} = \beta(w_j) \cdot E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j] - \alpha(w_j) \cdot b_{ij} + v_{ij}, \quad (4)$$

Thus, the employer's problem is:

$$\max_{i \in \{i': R_{i'j} \leq \phi_j\} \cup \{0\}} \{ \beta(w_j) \cdot E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j] - \alpha(w_j) \cdot b_{ij} + v_{ij} \}, \quad (5)$$

where $\eta_{0j} = b_{0j} = 0$ to represent that the outside option simply has value v_{0j} . Note that employers do not directly value worker observables x and only use them to form beliefs about match quality. The idea here is that ultimately employers only care about match quality, price, and idiosyncratic taste shocks.

Accordingly, each worker's problem is:

$$\max_{(b_{ij}, e_{ij}) \in [\underline{b}_j, \bar{b}_j] \times (0, \infty)} \left\{ \underbrace{P \left(i = \arg \max_{i' \in \{i'': R_{i''j} \leq \phi_j\} \cup \{0\}} \{ \beta(w_j) \cdot E[\eta_{i'j}|b_{i'j}, y_{i'j}, x_i, w_j, R_{i'j} \leq \phi_j] - \alpha(w_j) \cdot b_{i'j} + v_{i'j} \} \mid b_{ij}, e_{ij}, x_i, w_j, R_{ij} \leq \phi_j \right)}_{\text{Probability that } i \text{ is chosen conditional on } i \text{ being considered}} \times \underbrace{P(R_{ij} \leq \phi_j | x_i, w_j)}_{\text{Probability that } i \text{ is considered}} \times \underbrace{(b_{ij} - c_{ij})}_{\text{Net payment conditional on } i \text{ being chosen}} - \underbrace{\psi(e_{ij}, x_i, w_j)}_{\text{Upfront signaling cost}} \right\}. \quad (6)$$

Note that the probability that i is considered does not depend on i 's actions b_{ij} or e_{ij} . Thus, there is nothing any worker can do to affect her chance of being considered. She can only affect her chance of being chosen conditional on being considered. This fact is the key place where congestion enters the model. As the number of workers applying to each job increases, assuming firm attention constraints are held fixed, then the chance that any one worker is considered by the employer decreases. We will see in the next subsection that, in equilibrium, this effect will decrease workers' willingness to supply information to the market, i.e. they will lower their signaling efforts as congestion increases.

3.2. Equilibrium

We focus on symmetric pure strategy Bayes' Nash equilibria, which we assume to be unique. Workers will know their own cost type c_{ij} , their observables x_i , the project's observables w_j , the distributions of competitors $F_{X,W}$, costs F_c , match qualities F_η , signaling noise F_ϵ , taste shocks F_v , the random components of ranks cores F_ζ , and attention constraints F_ϕ – all of which are conditionally independent given x_i and w_j – the equilibrium strategy each worker of type (c, x) would play, and how the employer will compute $E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j]$.

The employer knows her own observables w_j , the realized set of applicants I_j , the observables x_i of each applicant i , her own taste shock v_{ij} for each applicant i , her outside option taste shock v_{0j} , her own realized attention constraint ϕ_j , the set of applicants for whom $R_{ij} \leq \phi_j$, the distributions of match qualities F_η and signaling errors F_ϵ , and the equilibrium strategy each worker of type (c, x) would play. She observes each worker i 's bid and signal (b_{ij}, y_{ij}) , and she computes $E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j]$ using Bayes' rule.

Assumption: The equilibrium choice of b is strictly increasing in c for all x and w .

Given this assumption, there will be a bijection between equilibrium bids and costs. Thus, when employers observe bids, they will be able to back out each worker's cost, and since they know each worker's equilibrium signaling effort choice as a function of c , x , and w , they will know the signaling effort that each worker chose in equilibrium, even though this effort is not directly observed.

Define $b^*(c, x, w)$ and $e^*(c, x, w)$ as the equilibrium bidding and effort functions, where $b^{*-1}(b, x, w)$, the inverse of $b^*(c, x, w)$, exists for every (x, w) according to the assumption above. Thus, when employers observe b_{ij} in equilibrium, they infer that worker i chose signaling effort $e^*(b^{*-1}(b_{ij}, x, w), x, w)$. We sometimes abuse notation and write this composite function as $e^*(b_{ij}, x, w)$.⁵

We can now compute $E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j]$, which can be rewritten as $E[\eta_{ij}|e^*(b_{ij}, x_i, w_j), y_{ij}, x_i, w_j]$. For readability, we will omit x_i , w_j , and the subscripts i and j from the following calculations, and we note that all of these calculations are done within discrete cells of x_i and w_j .

First, note that $y|e^*(b)$ is distributed normally since it is a linear combination of two independent normally distributed random variables η and ϵ . Also note that η and ϵ are independent of $e^*(b)$. Next, we show that $(y|e^*(b), \eta)$ is bivariate normal. For any $(p, q) \in \mathbb{R}^2$, $p \cdot y + q \cdot \eta = (p \cdot e^*(b) + q)\eta + p \cdot \epsilon$, which conditional on $e^*(b)$ is another linear combination of two independent normally distributed random variables. Thus, we can conclude $(y|e^*(b), \eta)$ is bivariate normal.

Using this fact, we can write

$$E[\eta|e^*(b), y] = m_\eta + \frac{E[\eta \cdot y|e^*(b)]}{\text{Var}(y|e^*(b))} (y - E[y|e^*(b)])$$

Note that since η and ϵ are independent, $E[\eta \cdot y|e^*(b)] = e^*(b) \cdot (V_\eta + m_\eta^2)$ and $\text{Var}(y|e^*(b)) = e^*(b)^2 \cdot V_\eta + V_\epsilon$. Moreover, since $E[\epsilon] = 0$, we can write $E[y|e^*(b)] = e^*(b) \cdot m_\eta$. Thus, putting back in x , w , and the subscripts i and j , and letting

$$A(b_{ij}, x_i, w_j) \equiv \frac{e^*(b_{ij}, x_i, w_j) \cdot (V_\eta(x_i, w_j) + m_\eta(x_i, w_j)^2)}{e^*(b_{ij}, x_i, w_j)^2 \cdot V_\eta(x_i, w_j) + V_\epsilon}$$

$$\text{and } B(b_{ij}, x_i, w_j) \equiv (1 - A(b_{ij}, x_i, w_j) \cdot e^*(b_{ij}, x_i, w_j)) \cdot m_\eta(x_i, w_j),$$

we can write:

$$E[\eta_{ij}|b_{ij}, y_{ij}, x_i, w_j] = A(b_{ij}, x_i, w_j) \cdot y_{ij} + B(b_{ij}, x_i, w_j). \quad (7)$$

⁵In future work, when measuring signaling effort, one will likely not observe e as a function b , x , and w , as one could observe multiple values of signaling effort for a given bid within a given cell of worker and employer observables. As a result, we propose that one assumes signaling effort is measured with exogenous error, so that the assumption that e is a well-defined function of b , x , and w in equilibrium can be maintained.

Note that A and B are functions of bids b_{ij} , observables x_i and w_j , and parameters $m_\eta(x_i, w_j)$, $V_\eta(x_i, w_j)$, and V_ϵ .

Thus, we can write out the employer's problem more directly:

$$\max_{i \in \{i' : R_{i'j} \leq \phi_j\} \cup \{0\}} \{ \beta(w_j) \cdot [A(b_{ij}, x_i, w_j) \cdot y_{ij} + B(b_{ij}, x_i, w_j)] - \alpha(w_j) \cdot b_{ij} + v_{ij} \}. \quad (8)$$

Note for the next subsection that we have now written out the employer's discrete choice problem as a function of data, parameters to be identified, and the taste shock v_{ij} .

Turning back now to the Freelancer's problem, we can conclude the equilibrium strategies clearly do not have an analytic solutions. However, we note that in equilibrium, and thus holding fixed the equilibrium strategies of all other players, we can write the ex-ante reduced-form probability that employer j considers worker i as $P^C(x_i, w_j)$, and we can write the ex-ante reduced-form equilibrium probability that employer j chooses (hires) worker i conditional on employer j considering worker i as a function of bid b and effort e as $P^H(b, e, x_i, w_j)$.

With these reduced-form equilibrium functions, we can rewrite the worker's problem in equilibrium as:

$$\max_{(b_{ij}, e_{ij}) \in [\underline{b}_j, \bar{b}_j] \times (0, \infty)} \left\{ P^H(b_{ij}, e_{ij}) \cdot P^C(x_i, w_j) \cdot (b_{ij} - c_{ij}) - \psi(e_{ij}, x_i, w_j) \right\}. \quad (9)$$

At the symmetric pure strategy Bayes' Nash equilibrium, the preceding problem has an equivalent solution to the original worker problem, and it produces equilibrium strategies $b^*(c, x, w)$ and $e^*(c, x, w)$ that together with all assumed distributions in the model will produce probabilities consistent with P^H . Note that since $P^C(x, w)$ does not take as inputs any equilibrium objects, and since we assume that all workers apply to jobs at which they arrive, $P^C(x, w)$ is an exogenous object to the equilibrium choices of bids and signal efforts.

We can now write down the two following first order conditions of the newly defined worker problem:

$$c_{ij} = b_{ij}^* + \frac{P^H(b_{ij}^*, e_{ij}^*, x_i, w_j)}{\left(\frac{\partial P^H(b, e, x_i, w_j)}{\partial b} \right) \Big|_{b_{ij}^*, e_{ij}^*}}; \quad (10)$$

$$\frac{\partial \psi(e, x_i, w_j)}{\partial e} \Big|_{b_{ij}^*, e_{ij}^*} = \left[\left(\frac{\partial P^H(b, e, x_i, w_j)}{\partial e} \right) \left(\frac{\partial P^H(b, e, x_i, w_j)}{\partial b} \right)^{-1} \right] \Big|_{b_{ij}^*, e_{ij}^*} \cdot P^H(b_{ij}^*, e_{ij}^*, x_i, w_j) \cdot P^C(x_i, w_j). \quad (11)$$

The first is the typical procurement auction first order condition that shows how workers mark down their bids from their costs. The second captures how workers trade off the benefit from increasing their signaling effort above what the employer expects with the cost of that effort increase.

We can think of congestion increasing as P^C decreasing for all workers, and at its limit, we can think of P^C approaching zero. As made clear from the second first order condition above, as P^C approaches zero, the right hand side of that equation approaches zero, meaning equilibrium choices of e must also approach zero, as per the assumptions on ψ . The logic here follows that of an all-pay auction. Since workers must pay for their signaling effort regardless of whether they are chosen, if they have a lower chance of even being considered in the first place, then they will be less willing to pay signaling costs up front. As all workers choose lower levels of signaling effort, due to the multiplicative structure of signal production, differences in signals between workers will become less informative as the signaling error will dominate match quality in signal production. In other words, as all workers provide less information to the market, employers will rely more and more on prior beliefs about match quality $m_\eta(x_i, w_j)$ to dictate choices. This effect can lead to significant ex-post hiring mistakes, and with low enough prior means of match quality for those workers likely to be in employers' consideration sets, employers will opt for the outside option more frequently, leading to inefficiently low match rates in the market.

We end this subsection by elucidating the logic behind the signaling equilibrium and the choice of signaling effort in the spirit of [Holmström \(1999\)](#). In general, the worker has an incentive to “fool” the employer by increasing effort above the equilibrium level. This incentive comes from the fact that increasing effort probabilistically increases the signal output without changing what effort level the employer assumes the worker chose, which leads the employer to increase her expectation of the worker's match quality, thereby increasing the worker's chance of being chosen. The equilibrium effort level, as described in equation (11), will thus be set such that any deviation upward will not be worth trading off against the increasing marginal cost of signaling effort.

Despite this incentive to “fool” the employer into thinking match quality is higher than it is, in equilibrium the employer will of course not be fooled and know exactly what the worker's effort level is. Accordingly, it may seem paradoxical that the worker would willingly expend costly effort on signaling. However, since in equilibrium the employer expects a certain effort level, the worker is also dissuaded from lowering her effort below this level, since doing so would, following the same logic as above, decrease her chance of being chosen.

In the words of [Holmström \(1999\)](#), “[T]he [worker] cannot fool the [employer]. Yet, he is trapped in supplying the equilibrium level [of effort] that is expected of him, because, as in a rat race, a lower supply of [effort] will bias the evaluation procedure against him.”

3.3. Identification

Although we leave estimation and empirical implementation of our model to future work, we present in this subsection the key identification arguments that show the model is indeed identified, and thus it is ripe for empirical application.

For ease of readability, we omit observables x_i and w_i as well as discussion of consideration sets from what follows. Before this omission, we discuss how one would incorporate these into identification arguments.

Since we are using discretely binned observables x and w , all of what follows can be thought of as being done within bins of $X \times W$. Of course, the arrival distribution $F_{X,W}$ itself can be identified by simply observing the realized distribution of observable types x that arrive at each job of observable type w . The attention constraint distribution F_ϕ can be identified simply by having data on ϕ for each job. We propose future work measures ϕ with click data from the employers showing how far down the ordered list of applications employers search.

The reduced-form consideration set probability P^C can be identified by observing the joint distribution of realized ranks R_{ij} , attention constraints ϕ_j , worker observables x_i , and employer observables w_j . Since the following analysis will be within bins of $X \times W$, P^C will just be a fixed probability. As a result, we simply define $P^*(b, e, x, w) = P^H(b, e, x, w) \cdot P^C(x, w)$ as the unconditional reduced-form probability of winning, and just write $P^*(b, e)$ below. Finally, we note that we can identify F_ζ by observing the rank scores r and observables x , such that we can identify $r(x_i)$.

Specifically, we are interested in identifying the distribution of match quality m_η and V_η , the distribution of signaling error V_ϵ , the distribution of costs F_c , the marginal cost function of signaling effort $\psi'(\cdot)$, and employer preferences β and α . We break up identification into separable parts that we suggest will mirror separable estimation routines in future work.

3.3.1. Signaling Model

First, we begin with the simplest part of the identification argument – the signaling portion of the model. We will observe the joint distribution of (y_{ij}, e_{ij}) , and recall that η_{ij} , e_{ij} , and ϵ_{ij} are all independent. Thus, $E[y_{ij}] = E[e_{ij}] \cdot E[\eta_{ij}] + E[\epsilon_{ij}] = E[e_{ij}] \cdot m_\eta$, and thus m_η is identified:

$$m_\eta = \frac{E[y_{ij}]}{E[e_{ij}]} \quad (12)$$

Next, note that $\text{Var}(y_{ij}) = \text{Var}(e_{ij} \cdot \eta_{ij}) + \text{Var}(\epsilon_{ij}) + 2\text{Cov}(e_{ij} \cdot \eta_{ij}, \epsilon_{ij}) = E[e_{ij}^2] \cdot V_\eta + \text{Var}(e_{ij}) \cdot m_\eta^2 + V_\epsilon$. Substituting $\frac{E[y_{ij}]}{E[e_{ij}]}$ for m_η and rearranging, we can conclude that we identify the following linear combination of V_η and V_ϵ :

$$E[e_{ij}^2]V_\eta + V_\epsilon = \text{Var}(y_{ij}) - \text{Var}(e_{ij}) \left(\frac{E[y_{ij}]}{E[e_{ij}]} \right)^2. \quad (13)$$

Moreover, note that $\text{Var}(y_{ij}|e^2) = e^2 \cdot V_\eta + V_\epsilon$, so putting this together with the last equation, we can see that both V_ϵ and V_η are identified.

The intuition behind these arguments follow the usual intuition from identifying linear models. Since ϵ_{ij} is mean zero, η_{ij} , e_{ij} , and ϵ_{ij} are independent, and we observe the means

of e_{ij} and y_{ij} , we identify the m_η just by taking means. Note that this makes m_η sensitive to the measurement of e_{ij} and y_{ij} . We avoid the importance of this sensitivity by letting β be a free parameter. In particular, the level of η only matters insofar as it factors into employer preferences, so if we arbitrarily doubled m_η , a correctly specified model would halve β .

3.3.2. Auction Model

With m_η , V_η , and V_ϵ identified, we move to the auction portion of the model to see how bidders' strategies can be used to identify the worker's costs and marginal signaling effort costs.

Ideally, we would follow a GPV-style (Guerre, Perrigne, and Vuong, 2000) argument applied to the workers' reduced-form first order conditions to recover F_c and $\psi'(\cdot)$. However, the first barrier is that unlike typical empirical auction environments, we do not have that the probability of winning is equal to the probability of having the lowest (or highest) type.

We fail to have this condition for two reasons. First, we have two dimensional types, one of which is unknown to the workers. So while it is the case that the worker with the lowest cost will bid the highest, she will not necessarily win, since she may not have the highest match quality. Second, even if a worker had the lowest cost and the highest match quality, she still may not win, given she may get an unlucky draw of signaling error lowering her signal and thus lowering the employer's belief about her match quality.

As such, we cannot analytically compute the probability of being chosen. However, since we observe the joint distribution of e_{ij} , y_{ij} , b_{ij} , and whether or not a worker is chosen, we can identify $P^*(b, e)$ at the current equilibrium evaluated at the observed combinations of (b_{ij}, e_{ij}) . These observed combinations will be the equilibrium pairs of (b_{ij}^*, e_{ij}^*) . Note that this does not solve our problem yet, since we do not just need to know $P^*(b, e)$ evaluated at equilibrium pairs, but also its two partial derivatives evaluated at those same pairs as per the workers' reduced-form first order conditions.

To identify these derivatives, we however need to identify $P^*(b, e)$ at all pairs of (b, e) in open sets around each observed (b_{ij}, e_{ij}) . Conveniently, our model allows us to do just that.

First, we note that $P(j \text{ chooses } i | b_{ij}, y_{ij}, e_{ij}) = P(j \text{ wins } i | b_{ij}, y_{ij})$. Since the employer only observes bids and signals, she will only choose her favorite application based on those two observables and her idiosyncratic shock v_{ij} , which is assumed to be independent of everything else. Thus, once we observe b_{ij} and y_{ij} , we do not gain any more information about worker i 's likelihood of being chosen by employer j by observing e_{ij} . We abuse notation and write $P^*(b, y)$ to denote the reduced-form probability of winning at the current equilibrium as a function of the chosen bid b and the realized signal y . Thus, we can now write:

$$P^*(b, e) = \int_{-\infty}^{\infty} P^*(b, y) f_{y|e}(y) dy, \quad (14)$$

where $f_{y|e}$ is the PDF of the distribution of y_{ij} conditional on e_{ij} . Moreover, note that once we condition on effort e_{ij} , the signal y_{ij} is determined only by two i.i.d. random variables, η_{ij} and ϵ_{ij} , so we only need joint data on (y_{ij}, e_{ij}) to identify $f_{y|e}$.

Next we claim that we can identify $P^*(b, y)$. Given η and ϵ have full support on the real line, so will $y_{ij}|e_{ij}$ for each choice of e . Thus, we will observe the full support of the joint distribution of b_{ij} , y_{ij} , and whether i wins j . As such, we can identify $P^*(b, y)$. The logic here is that given how e_{ij} randomly and exogenously translates to y_{ij} , even though we see a selected set of equilibrium pairs (b_{ij}, e_{ij}) , we see all pairs of (b_{ij}, y_{ij}) , and thus can use them to identify $P^*(b, y)$.

Putting the previous arguments together, we identify the right hand side of equation (14), and thus identify $P^*(b, e)$ over its entire domain, which assuming it is differentiable, allows us to identify its partial derivatives at all interior points.

The intuition behind this whole argument is as follows. Imagine a worker with type c deviates to an off-equilibrium bid, say b' , but maintains the same equilibrium effort choice for her type, say e . Let c' be the type of worker who chooses b' in equilibrium, and let e' be the equilibrium effort chosen by type c' workers. We thus observe the success of workers choosing (b', e') , yet we are trying to identify the counterfactual success of a worker who chooses (b', e) . Due to the randomness and full support of signals conditional on effort, we also observe the success of a worker whose ex-post bid equals (b', y) for all y . Thus, by re-weighting that success of ex-post choices (b', y) according to the resulting signal distribution conditional on effort choice e , rather than e' , we can recover the desired quantity.

Similarly, imagine a worker with type c maintaining b but deviating to some effort choice e' . We know the success of ex-post pairs (b, y) for all y , and so by the same re-weighting argument, we can recover the success of a counterfactual ex-ante choice (b, e') .

We are now ready to turn back to the workers' reduced-form first order conditions. Using the conjectured monotonicity of the equilibrium bidding strategy $b^*(c_{ij})$, assuming $P^*(b, e)$ is differentiable in both arguments, and assuming $\frac{P^*(b_{ij}^*, e_{ij}^*)}{\left(\frac{\partial P^*(b, e)}{\partial b}\right) \Big|_{b_{ij}^*, e_{ij}^*}}$ will be monotonic in

b_{ij} , we can follow GPV and write:

$$c_{ij} = b_{ij} + \frac{P^*(b_{ij}, e_{ij})}{\left(\frac{\partial P^*(b, e)}{\partial b}\right) \Big|_{b_{ij}, e_{ij}}}. \quad (15)$$

Therefore, for each observed (b_{ij}, e_{ij}) , we can use equation (15) to back out the cost type associated with each pair. This will allow us to non-parametrically identify F_c .

Similarly, we can write:

$$\psi'(e_{ij}) = P^*(b_{ij}, e_{ij}) \cdot \left[\left(\frac{\partial P^*(b, e)}{\partial e} \right) \left(\frac{\partial P^*(b, e)}{\partial b} \right)^{-1} \right] \Big|_{b_{ij}, e_{ij}}. \quad (16)$$

Therefore, for each observed (b_{ij}, e_{ij}) , we can use equation (16) to back out the marginal cost of signaling effort associated with each effort choice. Thus, we identify a differential equation in $\psi'(\cdot)$. Note that we cannot use these same arguments to identify the level of

$\psi(\cdot)$. Yet, since we assume $\psi(0) = 0$, we have a boundary condition to solve the differential equation identified above and recover $\psi(\cdot)$.

The intuition for how we use these first order conditions to identify F_c and $\psi'(\cdot)$ is straightforward. The level of bid each worker chooses relative to how the probability of winning responds to bidding identifies the underlying cost relative to which workers are bidding. And the level of effort each worker chooses relative to how the probability of winning responds to effort identifies how marginally costly it is to raise signaling effort.

3.3.3. Discrete Choice

At this point, all we have left to identify is β and α . Recall that the employer's problem, defined in problem (8), is a typical discrete choice problem over ex-post bids (b_{ij}, y_{ij}) where the objective function is linear in y_{ij} , β , and α and nonlinear in b_{ij} , m_{η} , V_{η} , and V_{ϵ} .

The relevant moments we can use here are how often the employer chooses the outside option, and which worker she chooses when she does choose an inside option. The former allows us to identify the levels of α and β relative to the normalized distribution of outside option values, F_v . The latter allows us to identify the difference between α and β , as per the usual discrete choice identification arguments.

Therefore, we have identified all the desired parameters, and are in fact over identified in that we can use the various levels of $\text{Var}(y_{ij}|e)$ for different values of e . We hope these arguments can also provide the framework for an estimation routine that follows the same split of the model into the signaling, auction, and discrete choice components.

4. Reduced-Form Evidence : Screening Costs and Mismatches in Digital Marketplaces

This section builds on our model of information provision in a two-sided matching market and provides empirical evidence that labor congestion in digital marketplaces can incentivize workers to send less informative signals of their quality to employers.

The section is split into two parts. First, we present reduced-form results that suggest that market congestion incentivizes workers to reduce the amount of efforts put in their work-quality "signal", hereby creating high information frictions about match quality. Second, we describe how we empirically build measures of writing effort and writing quality in order to quantify the effect of these variables in the matching process.

4.1. What Impedes Matching? Empirical Evidence of Market Congestion

This section presents evidence that are suggestive of a "bad congestion equilibrium" on international online labor platforms. The difficulty to search and be selected for jobs in such thick marketplaces encourages workers to submit many non-selective and low-quality bids.

Consequently, clients are overwhelmed by the volume and lack of customization of received job applications, rendering the effort required to discern the best matches greater than the potential value of the bids themselves. This imbalance prompts a rational yet unfortunate response: clients frequently opt not to hire anyone or make hasty choices with minimal scrutiny. In turn, freelancers, facing a low likelihood of their bids succeeding, default to the strategy that appears most rational under the circumstances: submitting a high volume of undifferentiated bids. This equilibrium results in a suboptimal market dynamic where neither freelancers' efforts in crafting quality bids nor clients' efforts in careful selection are appropriately rewarded.

4.1.1. Evidence of Market Congestion

Our data reveals a significant bottleneck in the online job marketplace, which we refer to as *market congestion*. This congestion arises when workers non-selectively submit a large number of low-quality bids. The lack of effective signaling mechanisms for match-value leads to an excess number of fast, un-customized bids, which clutters the marketplace and makes it difficult for good matches to occur.

Every job posted on the platform by prospective employers receives on average 25.4 bid applications (Figure 4) and the majority of these bids arrive within minutes after postings. The mean bid is submitted 4.9 hours after the job was posted, and the median bid is submitted within the first 9 minutes (Figure 5). By the end of the first day, almost all projects posted on the platform have received the totality of their bids.

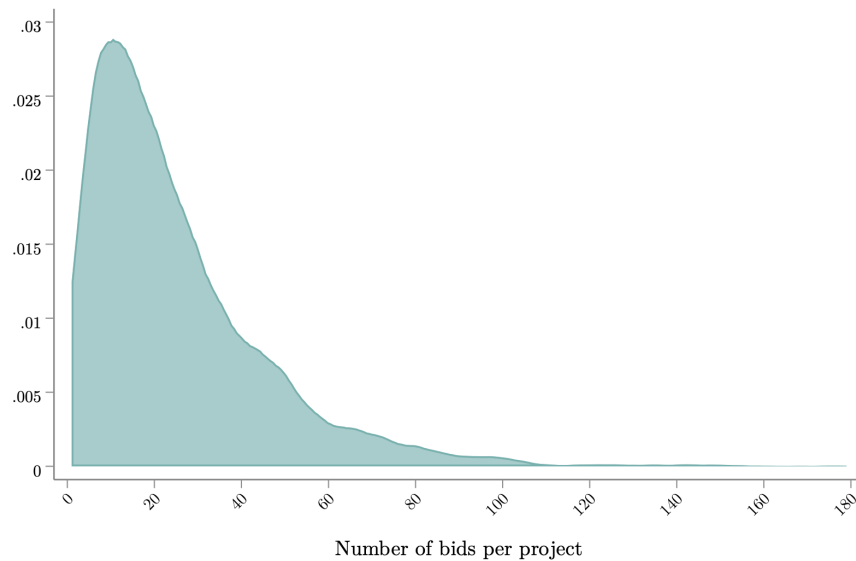


Figure 4: Number of Bids by Job Posting

Notes: This figure plots the distribution of the number of bids received for each job ("project") posted on the platform. Bids are posted by prospective workers and are composed of a monetary part (the bid amount offered to perform the outsourced task) and a written job application. Every job posted on the platform by prospective employers receives on average 25.4 bid applications, with a standard deviation of 22.4.

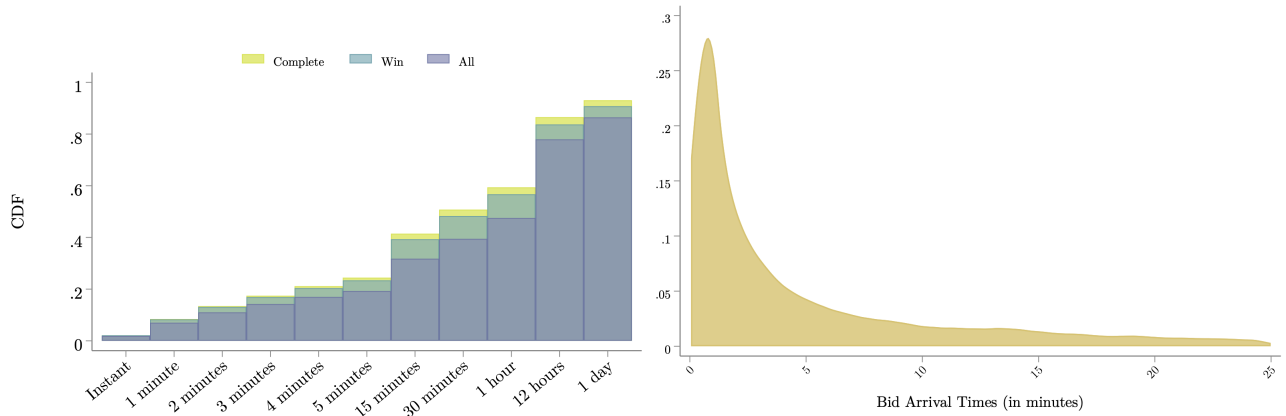


Figure 5: Large Volume of Bids Arrive Soon After Posting

Notes: These figures plot the distribution of bids arrival time, measured as the time between a job ad was posted by a prospective employer and the time a worker’s bid was submitted. The right figure plots the entire distribution of bid arrival times, underlining that most bids are submitted under 10 minutes. The mean bid is submitted 4.9 hours after the job was posted, and the median bid is submitted within the first 9 minutes. The left figure further breaks down the bid submission times into discrete bin, and displays the CDF of bid arrival times by project completion status. The purple bins are for all projects listed on the platform. The blue bins are for job postings that were awarded to a freelancer (i.e. a winner as chosen among all bids). The yellow bins are for projects that have been completed (i.e. the worker successfully finished the outsourced tasks). By the end of the first day, almost all projects posted on the platform have received the totality of their bids. Projects that are awarded to a worker and completed projects tend to display more bids arriving after 15-30mn.

Workers pay for membership plans that override the free-plan’s bid threshold and allow them to bid a lot (see Appendix C.5 for a detail of the plans). The distribution of bids made by workers every month evidence that the top percentiles of workers, as measured by the number of monthly bids submitted, bids above the 50 bids per month limit (Appendix Figures A.2 and A.3.)

4.1.2. Job Application Efforts Decrease with Congestion

With million of workers across the world competing for jobs and almost as many employers outsourcing job tasks every hour, searching and being selected for jobs on the platform is difficult. The thickness of the marketplace is one of its attractiveness, but its downside is that it can be very hard to find a match on Freelancer.com. In our data, around 90% of all freelancers who joined between 2016 and 2018 never find a job, despite submitting, on average, over 5 bids per month for 4.5 months before exiting the market. Similarly, around 62% of all posted projects between 2016 and 2019 never get matched to a worker, despite receiving on average 40 bids over this four-years period. While new prospective workers on the platform get less than 0.01 job per submitted bid, even experienced workers having already completed some jobs get on average 0.02 job per bid (Figure 6).

This situation is not unique to Freelancer.com; researchers estimate that out of the over 163 million workers who signed up for online freelancing, only 14 million of them have ever received work (Kässi, Lehdonvirta, and Stephany, 2021).

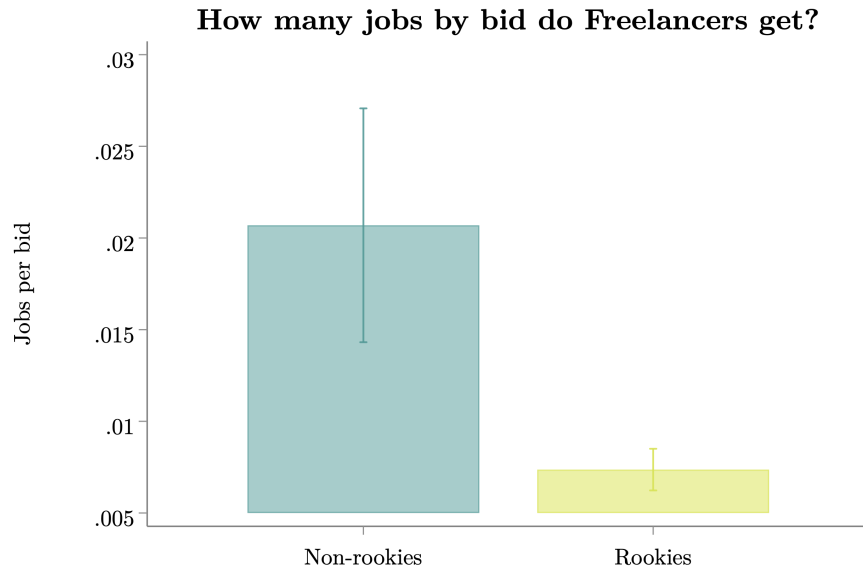


Figure 6: Getting a Job has Low Probability

Notes: This figure displays the ratio of jobs obtained per bid submitted across a 5% random sample of workers (“freelancers”) working on the platform between 2016 and 2019. The conversion rate from bid to job is extremely low, and even “non-rookies” get on average only 0.02 jobs per bid. Rookies are defined as all workers who are active on the platform but have not yet received a first milestone payment at the time of bidding. Non-rookies are workers that have received at least one payment at the time of bidding.

The low probability of winning incentivizes workers to submit bids quickly, often without customizing them to the specific jobs they apply for. The average worker bids extremely fast: they only take 2.6 minutes to write and submit their bid proposal (with a standard deviation 5.13). Figure 7 displays the distribution of bids time, measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid. Bids are composed of both a monetary amount and a written job application signaling quality and fit. If thus encompasses both reading time for the job posting and job proposal writing time. 50% of all workers submit their job application within one minute of opening a job posting. The bottom 25th percentile submits bids within 0.33 minute and the top 5% within 10 minutes.

Despite the fact most workers write and submit their bid under 3 minutes, there is some substantial differences between writing a job proposal for a few seconds, 1 minute and 3 minutes. There is a strongly positive relationship between the time workers spend to bid for a job and the length of the job description ad, as written by the prospective employer (Figure 8). This validates that the bidding time measure is a good predictor of the time and efforts it takes workers to write and submit their bid proposal. This bidding time measure includes the time it takes for the freelancer to read the job ad.

4.1.3. High Information Frictions Impact Match Rates

Employers receive a large volume of fastly written, copy-pasted bids within minutes of posting a job (Figure 10), making the process of selecting the best-fit worker arduous and

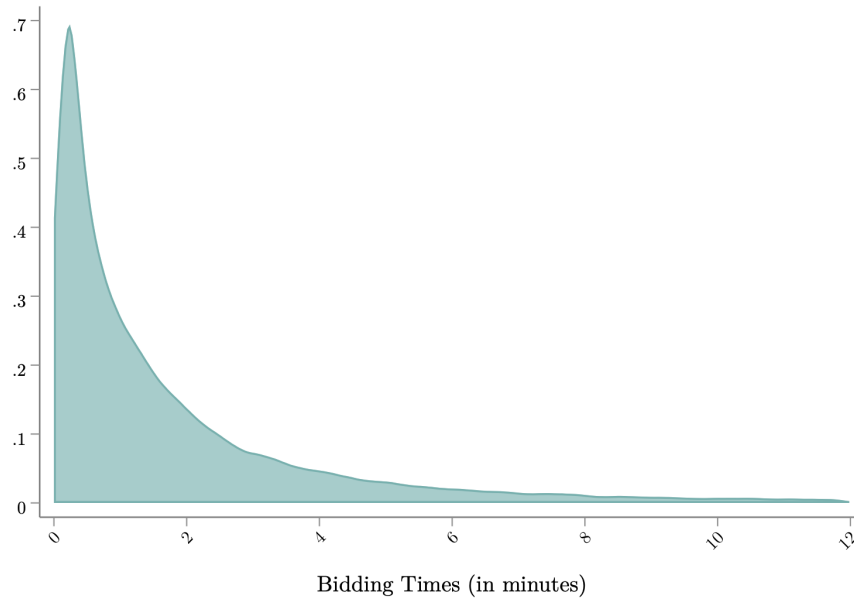


Figure 7: Workers Submit Job Applications Fast

Notes: This figure plots the distribution of bidding time, in minutes, for workers present on the platform in 2019. Bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid (composed of a monetary amount and a written job application signaling quality and fit). Bidding time thus encompasses both reading time for the job posting and job proposal writing time. On average, workers spend 2.6 minutes to submit their bid from the moment they click on a job posting (standard deviation 5.13). 50% of workers submit bids within 1 minute. The bottom 25th percentile submits bids within 0.33 minute and 95% of workers bid within 10 minutes.

time-consuming.

When employers cannot efficiently differentiate between high and low-quality bids due to their volume and lack of distinctiveness, the probability of a match decreases. The employer's original willingness to pay for an outsourced service is effectively reduced by a high "hassle cost" of selection a winner, which in some situations may make even the most attractive bids unappealing. The hassle cost of looking at every submitted bid promotes employers to often opt not to pick any worker or to make selections with limited attention by picking one of the high ranked bids.

As the platform ranking algorithm favors established workers with a good rating history, and these workers are those that tend to bid the fastest (see Figure 10), these high-ranked bids tend to be written hastily. The probability that an employer effectively outsources the posted job to a worker strongly decreases with the number of bids received (Figure 9). 60% of projects posted in 2019 did not get awarded to a worker.

4.1.4. Freelancers Learn to Bid Fast and More Over Time

The employer's hassle costs of sorting through a high volume of bids increases the probability of not hiring anyone or picking the first worker that appears on the platform. This mechanism further decreases the success rate per bid and perpetuates a system where the

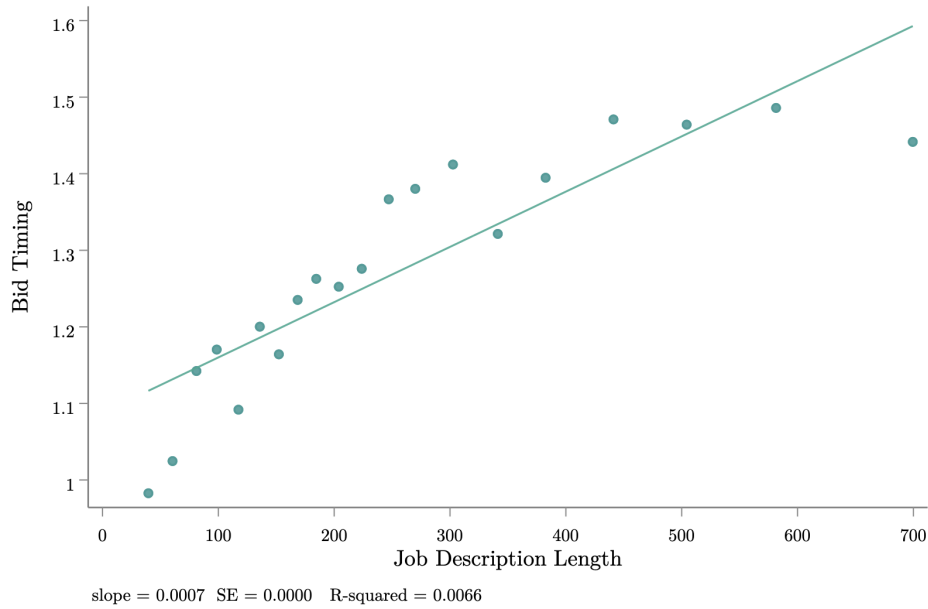


Figure 8: Job Posting Length and Bid Writing Time are Positively Correlated

Notes: This binscatter figure displays the relationship between the time workers spend to bid for a job and the length of the job description ad, as written by the prospective employer. The length of a job posting description is strongly positively correlated with longer bid writing time from the workers side. It validates that the bidding time measure is a good predictor of writing time efforts. The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. Even though most workers submit their bids under 2 minutes, time spent writing is tightly correlated with an employer project description length (measured in number of characters).

effort invested in producing high-quality bids is not justified by the returns. Slim chances of being selected creates a cycle where freelancers see little value in putting effort into their bids. Sending as many bids as possible becomes the strategy that provides the higher probability of getting a job from the worker’s point of view.

We find evidence that workers learn over time that the optimal strategy to obtain jobs in this congested marketplace is to bid quickly and non-selectively. First, “non-rookie”⁶ workers submit many more bids per months than “rookies” (Figure 11).

⁶Rookies are defined as all workers who are active on the platform but have not yet received a first milestone payment at the time of bidding. Non-rookies are workers that have received at least one payment at the time of bidding.

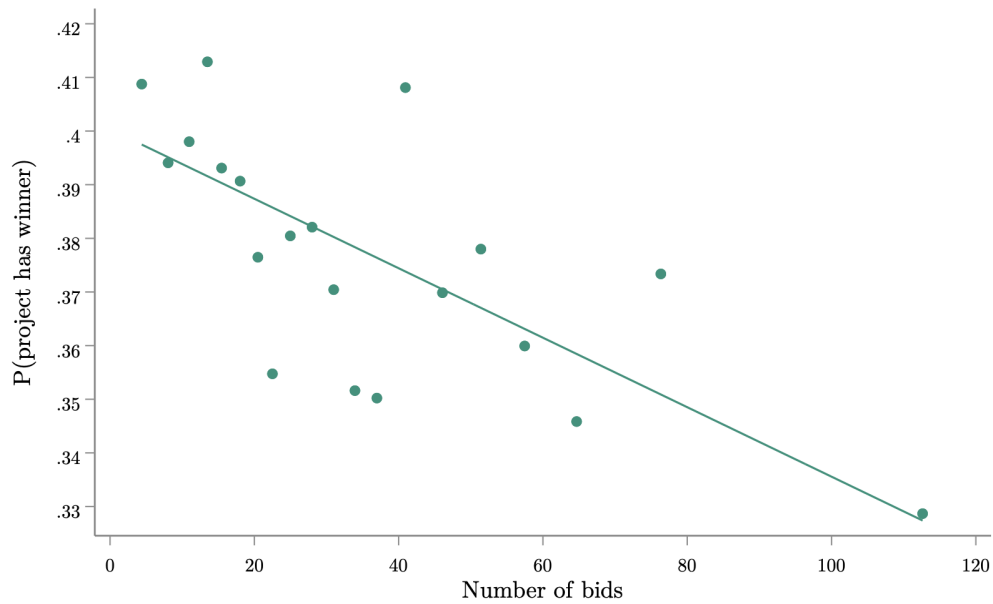


Figure 9: Clients Don't Pick Winners When Overwhelmed

Notes: This binscatter figure reports a strong negative correlation between the number of competitive bids submitted to a posted job and the probability that the prospective employer picks a winner. The average probability that an employer effectively outsources the posted job to a worker strongly decreases with the number of bids received. 60% of projects posted in 2019 did not get awarded to a worker.

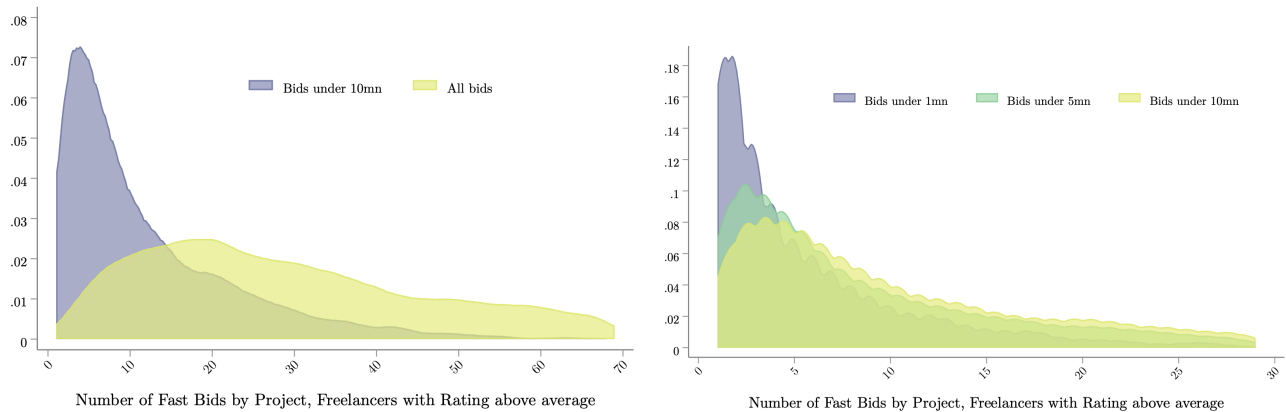


Figure 10: Clients Have Too Many Bids to Sort Through

Notes: These figures plot the distribution of arrival times for competitive bids sent to posted job offers. We focus on the sample of workers (“freelancers”) that display a star rating higher than average (“serious” workers). For each project posted on the platform between January 2016 and December 2019, we look at the number of bids received from serious freelancers, and the timing of those bids. The left figure plots the distribution of bids received under 10 minutes and compares it to all received bids. The right figure splits done fast bids from established freelancers into 3 categories: bids submitted under 1 minute, under 5 minutes and under 10 minutes. As the platform ranking algorithm favors established workers with a good rating history, and these workers are those that tend to bid the fastest, these high-ranked bids tend to be written hastily.

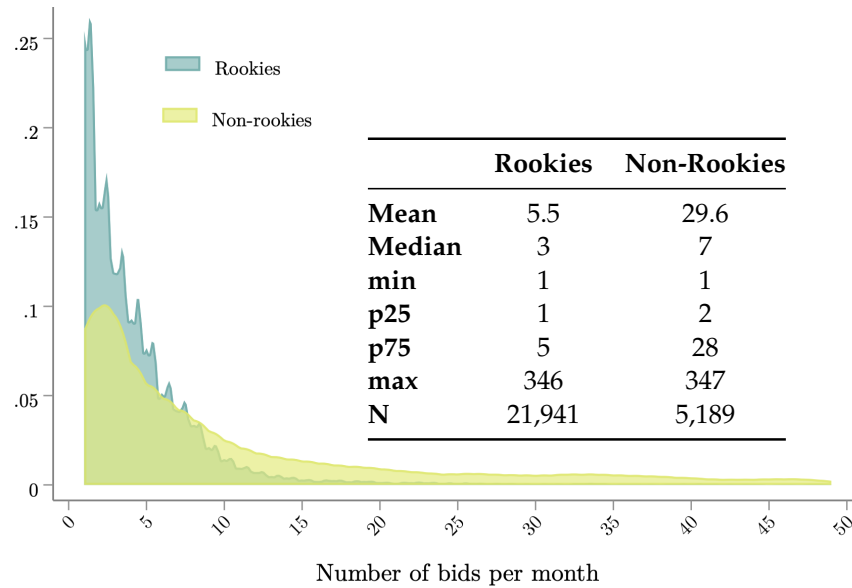


Figure 11: Experienced Workers Bid More

Notes: “Non-rookie” workers submit many more bids per months than “rookies”, thus supporting that workers learn over time that the optimal strategy to obtain jobs in this congested marketplace is to bid a lot. Rookies are defined as all workers who are active on the platform but have not yet received a first milestone payment at the time of bidding. Non-rookies are workers that have received at least one payment at the time of bidding.

We then look at two measures of workers seniority on the platform: i) the number of days they spent on the marketplace (as measured by the difference between the date they bid for a project and the first time they applied to a job) and ii) the number of reviews they received from past employers (measuring directly their past activity).

Firstly, the longer a freelancer has been working on the platform, the quicker they are applying to posted job ads (Figure 12). Looking at a random sample of active, non-rookie freelancers (i.e. freelancers having submitted bids to at least one projects in the last 6 months and having worked at least once), we find a strongly negative correlation between the number of days a worker has been active on the platform and the time they took to bid after the project was posted. This evidence that more experienced workers internalized that their chance of winning a job auction is higher if they bid shortly after the project was posted, as employers are more likely to view and consider their bids.

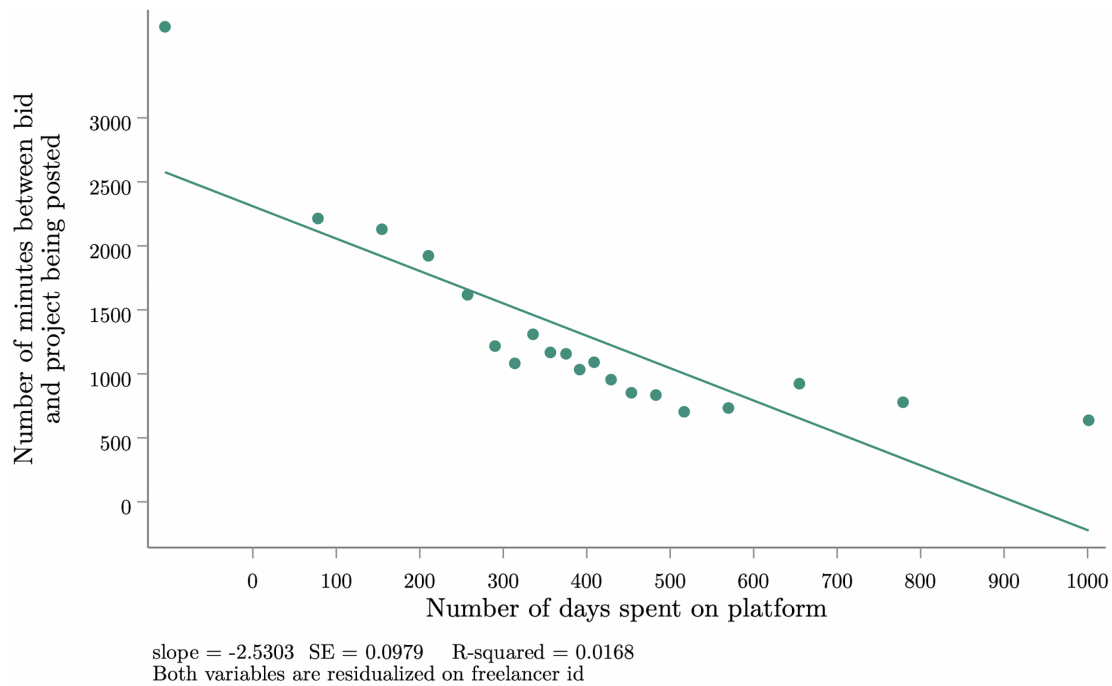


Figure 12: Timing of Bids Over Time

Notes: This binscatter figure shows a strongly negative correlation between the number of days a worker has been active on the platform and the time they took to bid after the project was posted. The longer a freelancer has been working on the platform, the quicker they are applying to posted job ads. The figure is based on a random sample of active, non-rookie freelancers working on the platform between January 2016 and December 2019 (i.e. freelancers having submitted bids to at least one projects in the last 6 months and having worked at least once). This evidence that more experienced workers internalized that their chance of winning a job auction is higher if they bid shortly after the project was posted, as employers are more likely to view and consider their bids.

Secondly, as a worker's observable characteristics become more predictive of a high match quality, workers can afford to spend less effort in sending a good signal (Figure 13). The number of reviews a freelancer received on the platform is strongly negatively correlated with the time they take at writing and submitting bids. Figure 13 emphasizes that as workers build an established history on the platform, they reduce the time they spend reading job postings, writing job proposals and submitting them (their "bidding time"). This suggests that workers internalize that employers will weight more heavily your already good prior as your observables become more predictive of high match quality, thus allowing you to spend less effort sending a good signal of quality and fit.

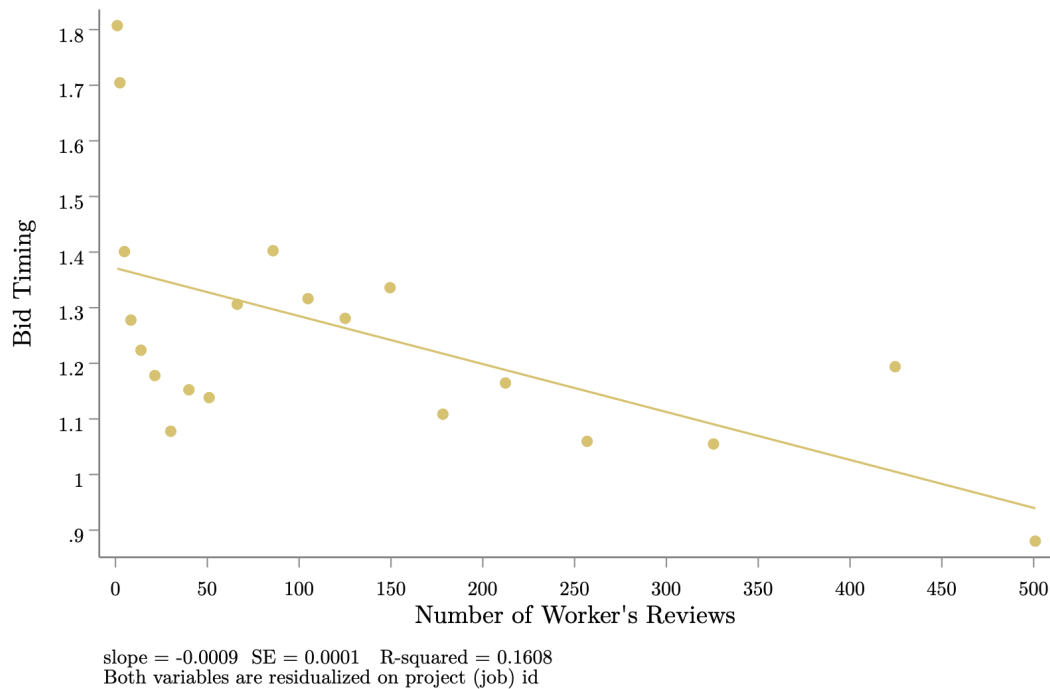


Figure 13: Workers Bid Faster as their Reputation Improves

Notes: This figure displays the cross-sectional relationship between bid timing and the number of worker’s reviews. The bidding time is measured as the timelapse between the moment the worker first clicked on a job posting and the moment they submit their bid application. The number of worker’s reviews captures the working history of a freelancer (more reviews means a more established history of work on the platform). A more established work history places workers at the forefront of bid rankings when they apply for jobs. This plot shows that workers bid faster as their reputation improves, and thus spend less time customizing their signal of quality and fit for the posted job. This is evidence that as an employer’s prior belief about a worker’s match quality goes up, the worker needs to spend less effort signaling during the application process.

4.1.5. Summary of Reduced-Form Evidence

Our empirical evidence suggests a bad congestion equilibrium in this digital marketplace for freelancer labor. For global workers trying to trade service tasks, the cost of investing high efforts in signaling “fit” for a job does not align with the expected returns. Conversely, as the time dedicated by worker to crafting job proposals decreases, the quality of their signals deteriorates, leading more employers to opt for alternatives outside the platform. The platform’s design, favoring workers’ history and past ratings in its bid-ranking algorithm, encourages a high quantity of low-quality, non-selective bids from freelancers, which in turn leads to a self-reinforcing equilibrium where employers are quickly overwhelmed by a volume of generic, copy-paste bids. Such an environment diminishes the return on investment for any single bid, discouraging efforts toward quality and customization. The market settles into an equilibrium where a lot of low-quality bids are submitted quickly and indiscriminately.

This pattern highlights the importance of information transmission in the matching process. It also underscores a critical decrease of signal quality amid competition, pointing

to the need for mechanisms that encourage more detailed and informative communication between workers and employers.

4.2. Measuring Writing Effort and Writing Quality

In order to quantify the effort a worker invests in tailoring their job application to a specific job post, that is their “signalling effort”, we construct a novel measure of writing quality that aims to reflect how well the written component of the bid aligns with the task description. To do so, we build two different sets of measures.

4.2.1. Measuring Methods: An Overview

NLP Methods. The first set relies on traditional Natural Language Processing (NLP) techniques, which focus on grammar accuracy or diction sophistication. We first measure the share of a worker’s job application material that is duplicated across applications. This first exercise provides an estimate of the extent to which a worker copy-paste their bid proposal across tasks and allows to measure the extent of time people actually spend writing. We then measure writing effort (assuming we are doing so with classical measurement error) by the time each freelancer spends writing their cover letter. We then rely on standard NLP measures of writing quality, which reduces to measuring textual complexity: how readable your text is (textual readability) and how rich it is (textual richness). Such approaches penalize misspellings, use of abbreviations etc.

New Approach: LLMs. The limitation of traditional NLP approaches is that they are little targeted to measuring “text fit”. To address this, we propose a new measure of writing quality that relies on a Large Language Model (LLM) to evaluate both the employer’s job descriptions and the workers’ job application materials submitted. Our measures aim to penalize job applications that, despite being well-written, are generic or recycled from other applications, and conversely, to reward letters that, albeit potentially less polished, directly address the job description’s specifics. This methodology is designed to assess the degree to which each proposal is tailored to and appropriate for the respective job posting.

This approach illustrates the innovative application of LLMs in quantifying aspects of text data that were previously beyond the reach of conventional analytical methods.

We implement and compare two different approaches to measuring the signaling fit of a worker’s job application utilizing the ChatGPT API.

4.2.2. LLM Strategy 1: Numerical Grading System

This first approach involves designing a prompt for ChatGPT to evaluate the writing quality of each bid proposal on a numerical scale. The scale will range from 0 to 10, where 0 indicates a proposal of poor quality that is likely generic, poorly written or irrelevant to the job posting, and 10 represents an excellent proposal that is highly customized, relevant, and demonstrates a clear understanding of the job requirements. This grading system allows for

a ranked assessment of writing quality and customization, providing a quantitative measure of each proposal’s alignment with the job description.

To ensure consistency and reliability in the grading process, the prompt to ChatGPT includes specific criteria for evaluation, such as relevance to the job posting, evidence of customization, clarity of expression, and how well the proposal addresses the job requirements. By applying this standardized assessment framework, we aim to generate a robust dataset of numerical quality scores for each bid proposal.

We perform various validation checks that our measure of writing quality is stable across multiple independent calls of the LLM and that it corresponds to more conventional, but less tailored, measures of writing quality.

4.2.3. LLM Strategy 2: Yes/No Evaluation with Posterior Probability Extraction

The second approach involves asking ChatGPT to determine whether a bid proposal appears well-suited and specifically tailored for the job posting through a simple yes/no question. This binary assessment will focus on evaluating the fit and customization of the proposal relative to the job requirements.

Upon receiving a yes/no answer from ChatGPT, we extract the model’s posterior probability of a “yes” response, which will serve as the score for each proposal. This score reflects ChatGPT’s confidence in the proposal’s suitability and customization for the job. To measure the posterior probability of a “yes” answer, we leverage ChatGPT’s ability to assess the likelihood of certain outcomes based on the input data. This process involves querying ChatGPT with a prompt that includes the bid proposal and the job description, asking it to evaluate the suitability and customization of the proposal in a yes/no format. ChatGPT, based on its internal models and the training it has received, can evaluate the input data and provide a yes/no answer. The API endpoint supports returning probabilities for its responses and this measure can directly be requested along with the yes/no answer.

The extracted posterior probability for a “yes” answer then serves as a quantifiable measure of the proposal’s alignment with the job description. This numerical score can be used to rank proposals, analyze the effectiveness of the bidding process, and potentially guide improvements in how freelancers craft their proposals.

The advantage of this method lies in its simplicity and the direct measure of fit and qualification it provides, offering a complementary perspective to the numerical grading system.

4.2.4. Implementation and Data Analysis

Note for the reader: this part is currently under implementation. First results should be out soon and will be used to empirically estimate the model detailed in Section 3.

For both LLM strategies, we systematically send text from freelancers’ bid proposals alongside the corresponding job descriptions to the ChatGPT API, using carefully designed prompts to elicit the desired evaluations. By employing these two approaches concurrently, we aim to capture both the nuanced differences in writing quality and the binary assessment

of fit and customization, thereby providing robustness tests of how well proposals meet job postings' requirements.

Our analysis will compare and contrast the outcomes of these strategies to determine the most effective method for measuring proposal quality in the context of online job platforms. Through this innovative use of LLMs, we anticipate contributing valuable insights into the optimization of matchmaking in two-sided markets, with implications for platform design and freelance labor dynamics.

We feed each task description to the LLM and prompt it to role-play as the firm posting the task. We then feed each cover letter (without any other information) to the LLM and prompt it to score the writing based on a judgement of how well the freelancer writing the cover letter will do on the task described. We perform various validation checks that our measure of writing quality is stable across multiple independent calls of the LLM and that it corresponds to more conventional, but less tailored, measures of writing quality.

5. Conclusion

In progress

In the rapidly evolving landscape of online labor markets, the process by which employers and workers find each other is pivotal. The efficiency of these matches has far-reaching implications, not just for the immediate parties involved, but for the broader digital economy. This paper investigate the dynamics of task-based online labor markets, focusing on how screening costs and information frictions impact job matching on these platforms. We use data from a major international digital marketplace to highlight the mechanisms of matching.

We develop a model of communication in a two-sided matching market with transfers. The model embeds a career concerns model of signaling into a scoring procurement auction in which buyers care both about match quality and price.

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Appendices

A. Market Description - Additional Reduced-form results

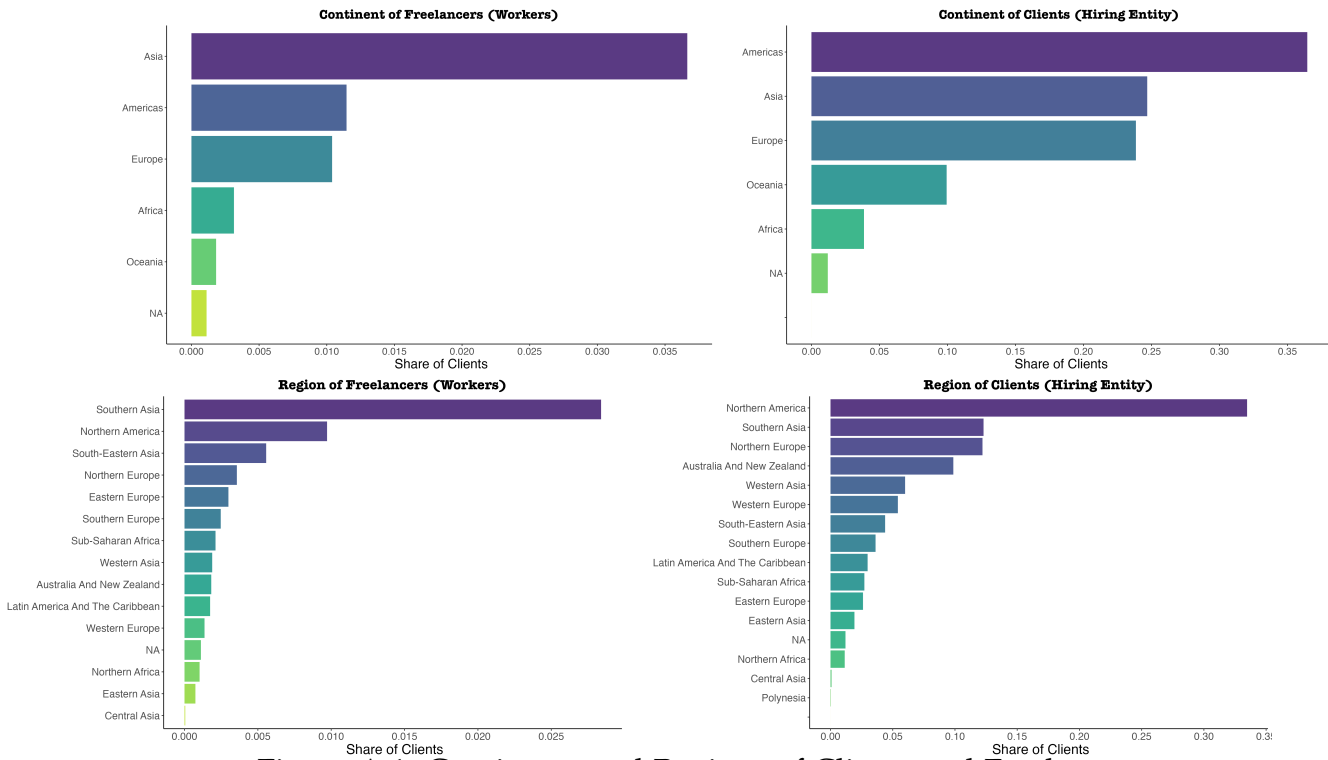


Figure A.1: Continents and Regions of Clients and Freelancers

Notes: These figures plot the share of workers (“freelancers”) and employers (“clients”) by continent and world regions. The two top figures show the geographical distribution of workers (left) and employers (right) by continent, while the two bottom figures show their spatial location across regions. Firms on the platform are largely from the Global North, while freelancers on the platform are mostly from the Global South. Most freelancers come from Asia, and more particularly Southern Asia, while most clients looking to outsource service tasks are from Northern America and Europe.

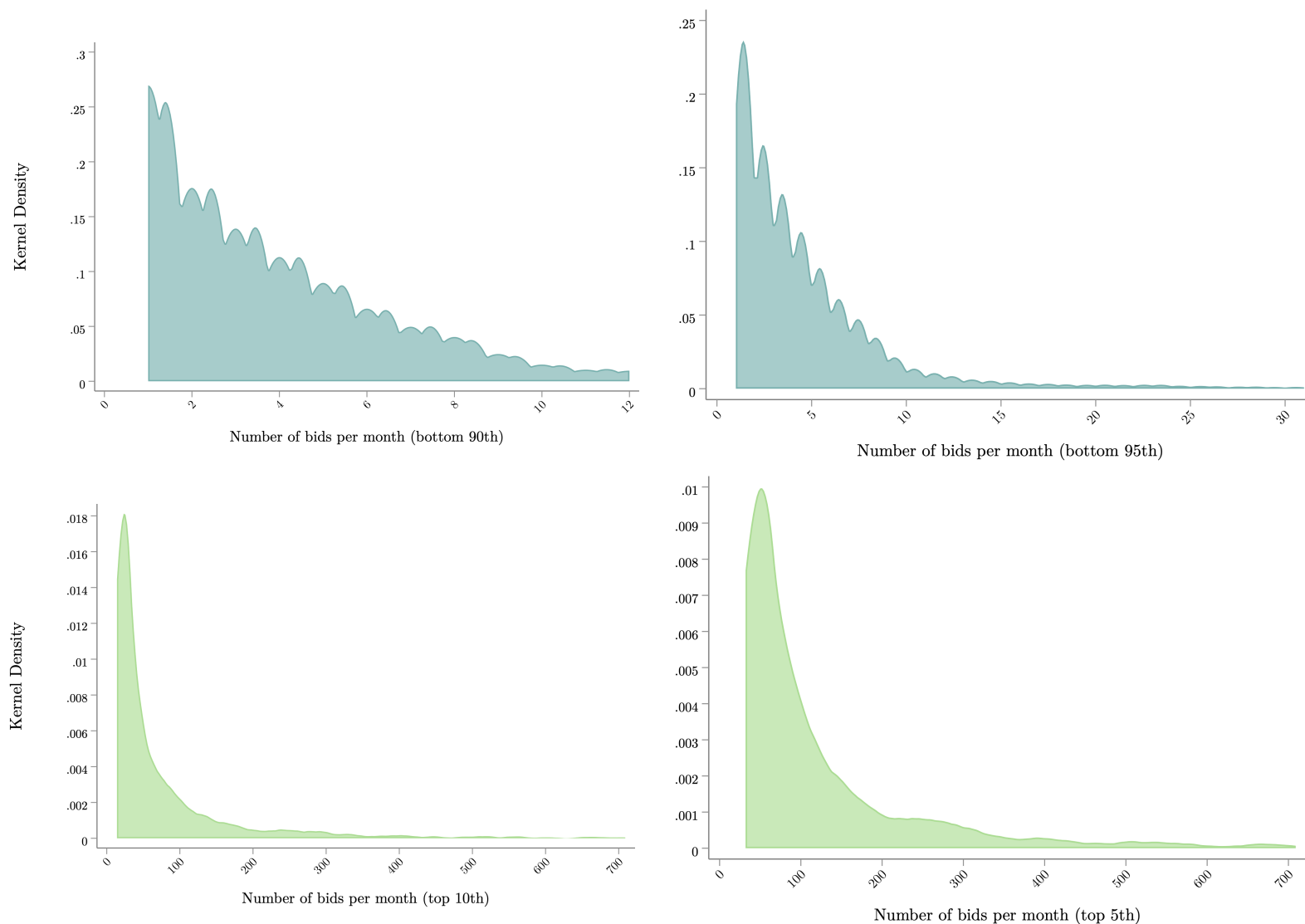


Figure A.2: Freelancers Are Paying for Memberships with High Bid Budgets

Notes: These figures plot the distribution of bids submitted by workers on the platform. The top two figures display the bottom 90th and 95th percentiles of workers, while the two bottom figures focus on the top 10th and 5th percentiles. Distributions are over workers. Workers at the top bid a lot. This provides evidence that workers pay for membership plans that override the free-plan's bid threshold and allow them to bid a lot (see Appendix A.3 for a detail of the plans).

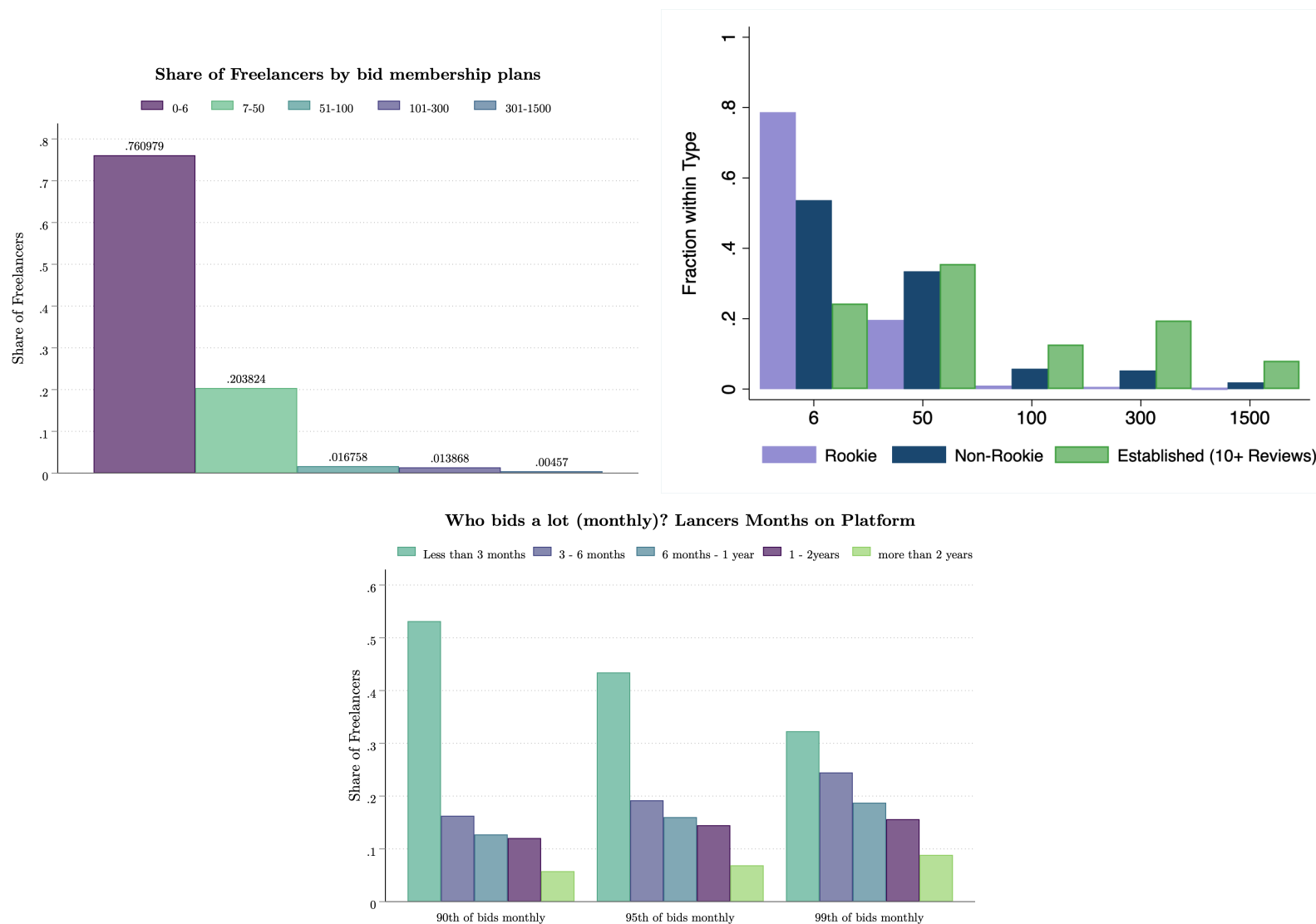


Figure A.3: Workers are Paying for Memberships that Permit to Submit more Bids

Notes: These figures display the distribution of workers (“freelancers”) across bid membership plans (see Appendix C.5 for a detail of the plans). Memberships allow workers to bid more than the free membership plan. There are currently 6 membership plans. The free membership plans allows to submit 6 bids per month, the second tier up to 50, and the three following plans up to 100, 300 and 1500. The top left figure shows that 20% of freelancers select to pay and submit up to 50 bids a month. Higher limit plans display smaller share of workers. The top right figure further decomposes membership payments by worker seniority type. “Rookies”, i.e. freelancers that never received a payment for a job, predominantly select into the first two tiers of memberships. More experienced freelancers tend to pay for higher limit plans. The more established the freelancers, the more likely they are to pay for plans that provide very high bid thresholds (including the 300 monthly bid and 1500 bid plans). The bottom figure further splits freelancer by the length of time they spent on the platform, and look at the number of bids they submit every month. The longer freelancers have been working on the platform, the more likely they are to bid a lot.

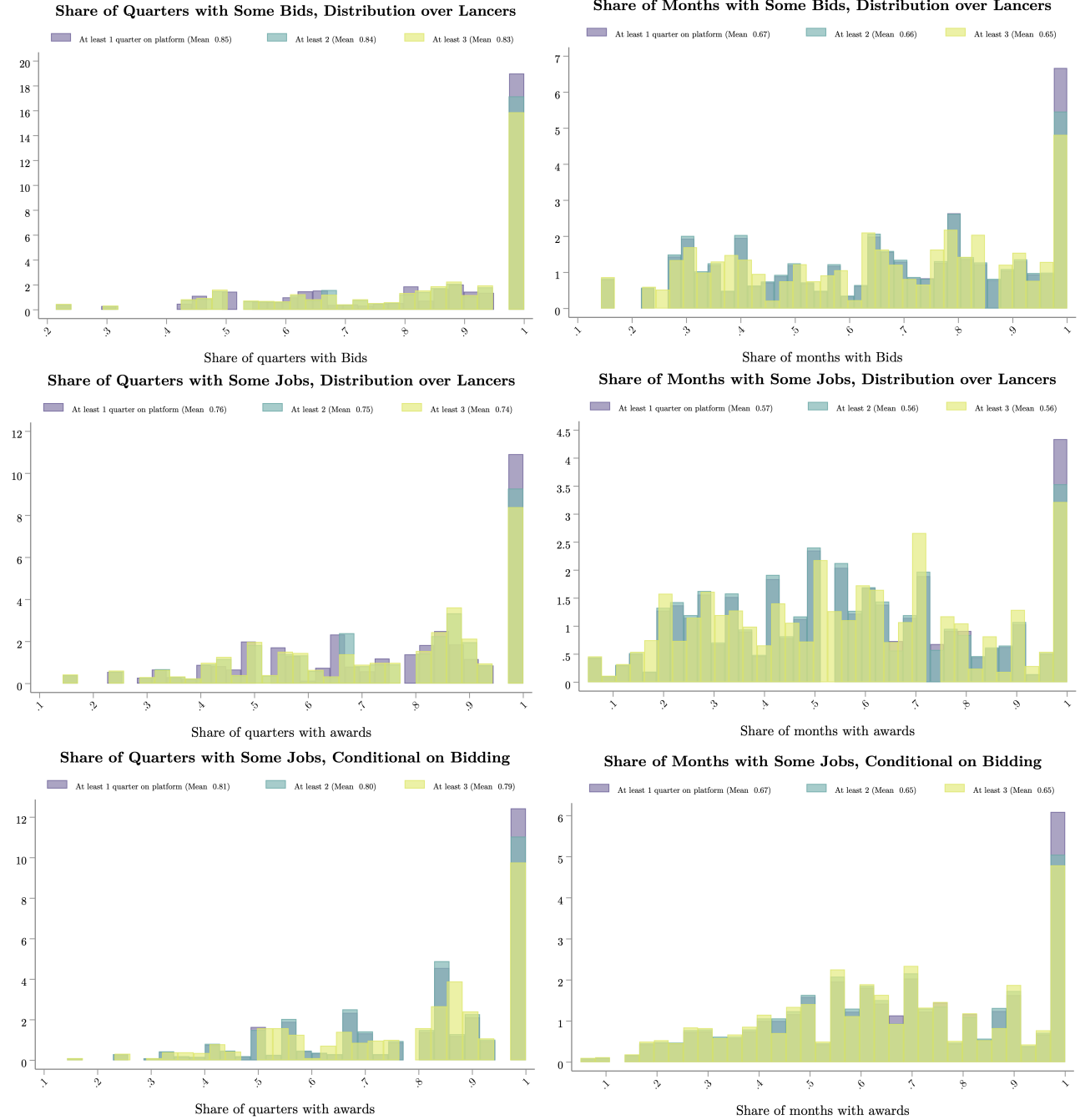


Figure A.4: Density Plots - Time Distribution of Bids and Jobs

Notes: These barplots display the time distribution of bids and jobs for workers (“lancers”) on the platform over the period january 2016 to december 2019. All figures on the left are at the quarter level and those on the right are at the monthly level. We split the bins for three groups of freelancers: i) those who have been on the platform for at least 1 quarter (in purple), ii) at least 2 quarters (green) and iii) at least 3 quarters. The first two top graphs plots the share of time during which freelancers submit some bids. The middle two graphs show the share of time during which freelancers have some jobs on the platform. The last two graphs show the share of time during which freelancers have some jobs on the platform, but conditional on bidding.

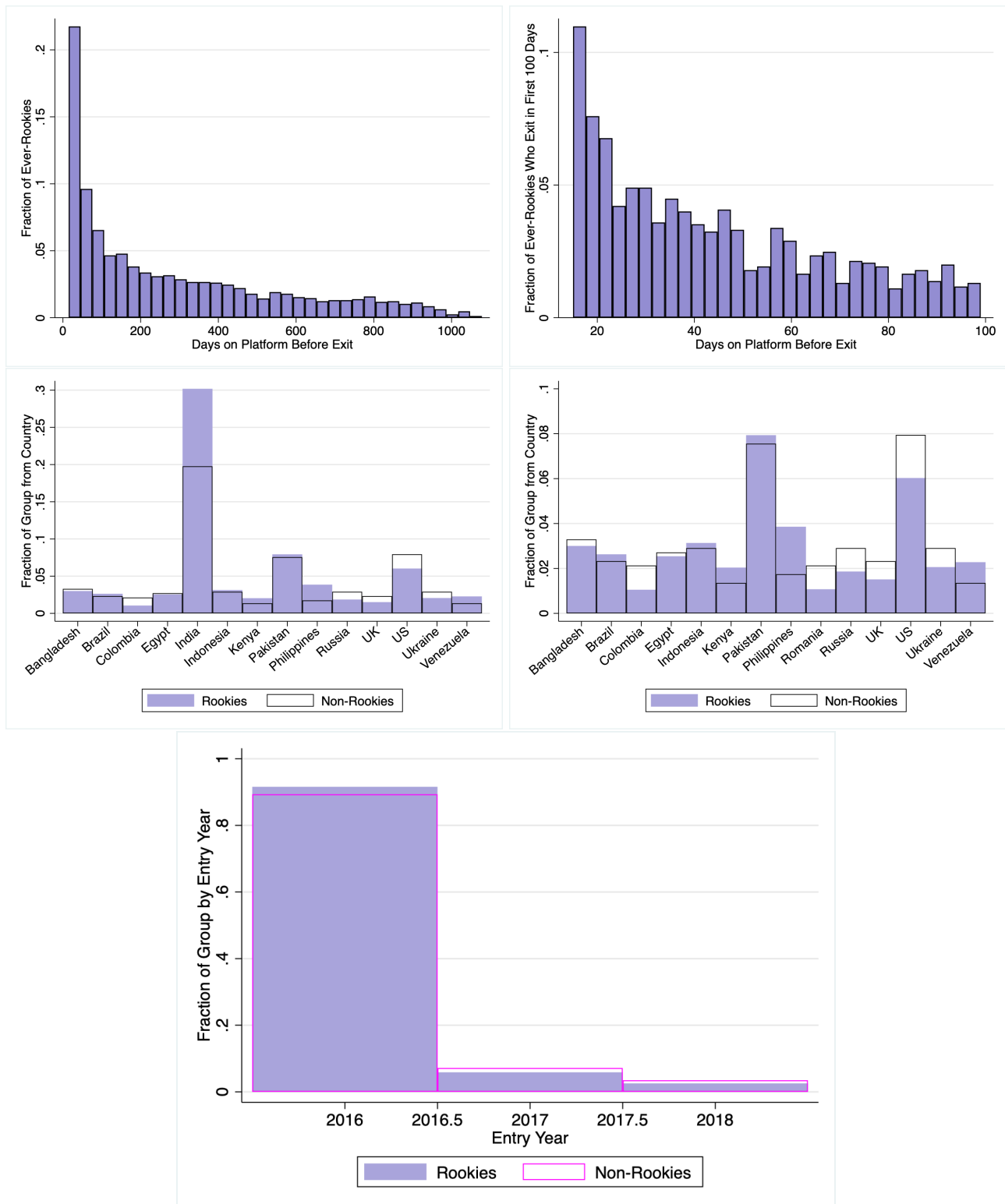


Figure A.5

Notes:

B. Data Appendix

The following list details the main variables we use from `Freelancer.com` for this project.

Identifiers include:

- Project ID
- Bid ID
- Client ID
- Freelancer ID

Client/Freelancer Information:

- Did the freelancer pay for upgrade
- Is the client/freelancer a new or returning user (Yes/No)
- Client's number of past projects
- Freelancer's number of past bids
- Number of past reviews and corresponding scores (for client/freelancer)
- Country and state (for clients/freelancers)
- Client's past total spend

Bidding Information:

- Bid timestamps
- Bid price
- Bid text
- Actions related to the bid: whether it was selected, at what price, final payment, reviews

Project information:

- Project upgrade fees charged
- Project milestones
- Project skills
- Milestone actions (release, reject, update, cancel) and timestamps
- Project budget with respective currency noted (or all in US dollars if possible)
- Revenue earned per project
- Indicators for which bidders clients chat with, including timestamps and character counts.
- Project success metrics and feedback provided.

C. Platform Details Appendix

Table C.1: Job Categories and Skills

Job Category	Skill Examples	Number of Skills
Websites, IT & Software	PHP, HTML, JavaScript, Software Architecture	15283
Design, Media & Architecture	Graphic Design, Website Design, Photoshop	15244
Mobile Phones & Computing	Mobile App Development, Android, iPhone	1752
Data Entry & Admin	Data Entry, Excel, Data Processing	1698
Engineering & Science	AutoCAD, CAD/CAM, Engineering	2394
Freight Shipping & Transportation	Dropshipping, Containerization, Import/Export	38
Business, Accounting, Human Resources & Legal	Business Analysis, Accounting, Finance	1317
Product Sourcing & Manufacturing	Product Design, Product Sourcing, Supplier Sourcing	161
Sales & Marketing	Internet Marketing, Social Media Marketing, Sales	3570
Translation & Languages	English (US) Translator, Spanish Translator, French Translator	574
Writing & Content	Content Writing, Article Writing, Copywriting	3858
Artificial Intelligence	Large Language Models (LLMs), Bot Development	136
Education & Tutoring	English Tutoring, Language Tutoring, Chemistry Tutoring	28
Health & Medicine	Food Safety, Counseling Psychology, Public Health	8
Telecommunications	Physical Site Survey, Active Site Survey, Pre-inspection visits	20
Jobs for Anyone	Virtual Assistant, Local Job, Travel Ready	256

Notes: This table summarizes the main job categories available on Freelancer.com. The second column provides examples of the type of skills workers can list on their profile and employers can link to projects. The third column summarizes the number of individual skills available under each broader job category. See <https://www.freelancer.com/job/> for a full list of skills.

My recent searches

Filter by:

Budget

☐ Fixed Price Projects

min

to

max

☐ Hourly Projects

min

to

max

All Durations

☐ Contests

min

to

max

Type

☐ Local Jobs
☐ Featured Jobs
☐ Recruiter Jobs
☐ Full Time Jobs

Skills

☐ .NET 5.0/6
☐ .NET
☐ PHP
☐ HTML
☐ CSS

Or enter skills

Languages

☐ English
☐ Spanish
☐ German
☐ French

Newest first

5,000 jobs found, pricing in USD

1

2

3

E-Commerce UI Design Across Platforms

6 days left

\$167 Avg Bid

21 bids

I'm in need of a UI designer who is skilled in creating Corporate and professional-looking web and mobile interfaces for an e-commerce platform. Key responsibilities include: - Developing a user interface compatible with Web, iOS and Android platforms. - Incorporating a sophisticated, corporate and professional design style. - Ensuring that the design promotes user-friendly navigation for online shopping. Ideal candidates should: - Have prior experience in creating UI design across multiple platforms. - Hav.

Graphic Design iPhone Mobile App Development User Interface / IA Website Design

need to made a indicator in esignal charting software

6 days left

VERIFIED

\$72 - \$78

0 bids

I'm looking for a software developer proficient in eSignal and TradingView applications. You will work on creating a specific indicator in eSignal charting software using the logic of a tradingview indicator Key skills and experience: - eSignal charting software - TradingView platform - Understanding of trading and stock market If you possess these skills, this could be the perfect project for you.

C Programming C++ Programming JavaScript Pine Script Python

Instagram Promotion for Brunch/Workshop Event

23 hours left

VERIFIED

\$111 Avg Bid

11 bids

I need a savvy Instagram marketer to promote a brunch/workshop I'm hosting. This event is open to both public and private attendees. Your main goal will be to increase event attendance. Ideal skills include: - Experience with Instagram marketing - A knack for engaging event promotion - Understanding of public and private event marketing.

Facebook Marketing Internet Marketing Marketing Social Media Marketing Twitter

URGENT

Bulk Certificate Generation with LaTeX and CSV Integration -- 2

6 days left

\$7 Avg Bid

2 bids

I am in need of a talented freelancer who can assist me in automating the bulk generation of certificates using LaTeX templates and CSV data integration. The project entails developing a streamlined process to efficiently produce certificates for events, workshops, and academic achievements. Key Requirements: 1. LaTeX Proficiency: The freelancer should possess advanced skills in LaTeX typesetting to ensure precise formatting and layout consistency across certificates. 2. CSV Data Integration: Ability to...

Data Entry Excel LaTeX

Amazon, eBay and online Dropshipping Expert

6 days left

VERIFIED

\$190 Avg Bid

13 bids

I am currently seeking an experienced dropshipping expert well-versed in product research and sourcing for Amazon and eBay marketplaces. The primary focus of this role will be on electronics. Key Responsibilities: - Research and source high-demand, profitable electronics for the dropshipping business - Analyze market trends and customer reviews to identify potential products - Stay updated with the latest trends in the electronics market Ideal Skills and Experience: - Previous experience in dropshipping,...

Figure C.1: Looking for Jobs - Platform Overview

Notes:

Freelancers For Hire

× Online Users

What work do you require?

Select a category of work

Skills

[Browse skills](#)

Countries

Specific Location

Exams

Hourly Rate (USD)

Rating

★ ★ ★ ★ ★

Online

☒ Online freelancers only (for instant chat)

1 2 3 4 5

Showing 9315 results

mikehurley

Expert SEO-Backlinks-Local SEO/GMB/Google Ad/SMM

4.9 ★★★★★ 10.0 \$ 6609 reviews \$10 USD per hour

SEO, Link Building, Internet Marketing, Marketing, Advertising

Highest 6600+ Reviews, 2000+ satisfied clients Glowing Reviews! Elevate Your Business with Expertise in SEO, Google Ads, and Facebook Ads! # Your digital journey deserves the best! Partner with the most reviewed and skilled SEO, Digital Marketing, and Website Development team of Elite Information Tech for...

seo4quality

Google #1 Rank | Google Ads | White Hat SEO

4.8 ★★★★★ 9.6 \$ 1564 reviews \$3 USD per hour

SEO, Internet Marketing, Link Building, Marketing, Search Engine Marketing

★ A Group of Professional Digital Marketing Experts with Google Certified ★ Quick Website SEO Offering the advance level SEO and Google ads service for our all clients to get the good result in terms of both organic and paid way. We have worked on different niche websites. We can perform the best practices for your...

dreamdezine

Where imagination & art meet.

4.9 ★★★★★ 9.2 \$ 3012 reviews \$25 USD per hour

Graphic Design, Photoshop, Logo Design, Illustrator, Website Design

Welcome to Dream Dezine, your one-stop destination for top-quality graphic design solutions tailored to elevate your business. Specializing in a diverse array of services including logo design, brochure creation, flyer and poster design, business card development, infographics, eye-catching PowerPoint presentations,...

xyzseo

SEO|SMM|Google Ads|Wordpress|Shopify|Graphic|Video

4.8 ★★★★★ 10.0 \$ 2652 reviews \$5 USD per hour

SEO, Internet Marketing, Link Building, Search Engine Marketing, Google Adwords

♦ Digital marketers who make your online brand visibility impressive ♦ White Hat SEO | Google AdWords | Social Media marketing | Web Development We offer complete digital marketing services. We have experience of 12+ years Complete Search Engine Optimization (SEO) Social Media Optimization (SMO) Google...

mhafujmahmud

Data Entry/Excel/Web Research/Product Listing/VA

5.0 ★★★★★ 10.0 \$ 202 reviews \$8 USD per hour

Data Entry, Web Search, Research, Search Engine Marketing, Excel

I am a professional freelancer committed to delivering reliable work with 100% accuracy according to my clients' needs. I am detail-oriented and work diligently to meet deadlines. My goal is to offer flexibility, enabling me to explore a variety of jobs that will enhance my skills and share them with future clients....

Figure C.2: Looking for Workers - Platform Overview

Notes: Employers looking for a worker to complete a project can either publicly post a project and open it for competitive bidding, or they can browse through freelancer profiles and directly send a request for hiring to a specific worker. However, the posting of a fixed-price project is the vastly predominant mode by which employers look for workers. The “Freelancers for Hire” section was not a prevalent feature of the platform in 2019-2020.

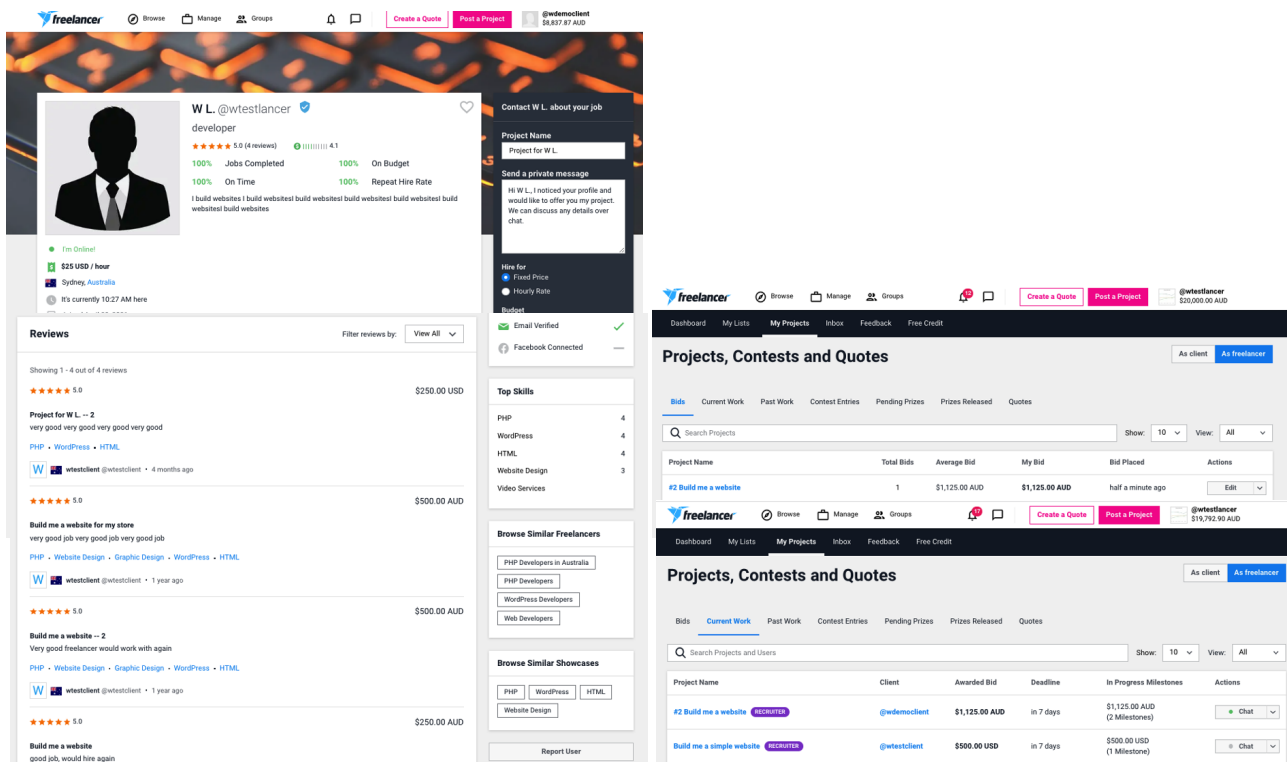


Figure C.3: Worker Profile

Notes:

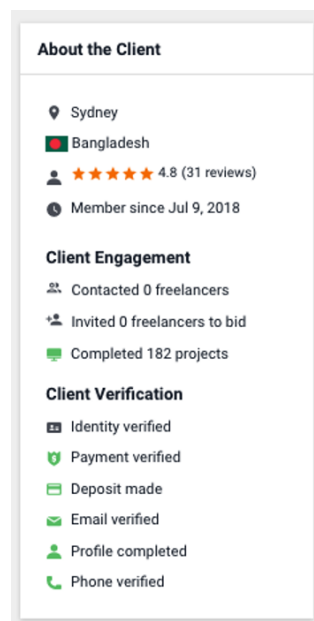


Figure C.4: Employers Profile (from Workers' Point of View)

Notes:

Try a Freelancer Membership

Designed to maximise your freelancer success and earnings! Save up to 20% on annual plans. Change plans anytime, conditions apply see FAQ.

Monthly Plan
Annual Plan

Basic

\$4.99

per month

Get Started!

50 Bids Per Month

50 Skills

Plus

FREE

~~\$8.95~~

1 month trial

Start Free Trial!

100 Bids Per Month

80 Skills

Professional

\$49.00

per month

Get Started!

300 Bids Per Month

100 Skills

Premier

\$99.00

per month

Get Started!

1500 Bids Per Month

400 Skills

Confirm release of payment for W L.



Build the fronted
\$525.00 AUD

Please ensure that you are satisfied with the work W L. has submitted, as Milestone Payments cannot be returned once released.

☒ My freelancer has delivered satisfactory work.

Cancel
Release Payment

Milestone Payments

Request Milestone

Created Milestones

Date	Description	Status	Amount	
27 Jun 2022	Build the fronted	Release Requested	\$525.00 AUD	Cancel
27 Jun 2022	Build the backend	In Progress	\$600.00 AUD	Request Release

Figure C.5: Memberships and Payments

Notes: